

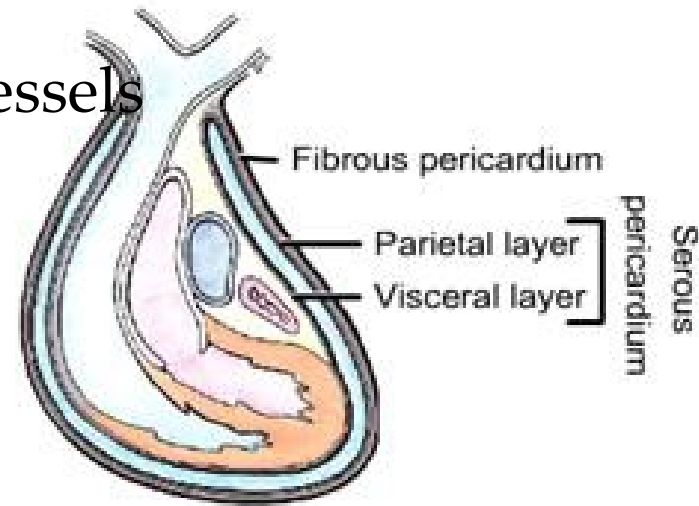
ANATOMY OF HEART

Pericardium

- The **middle mediastinum** includes the pericardium, heart, and roots of its great vessels
- Is double-walled **fibro-serous** sac
- Encloses the heart and roots of great vessels

Has two parts:

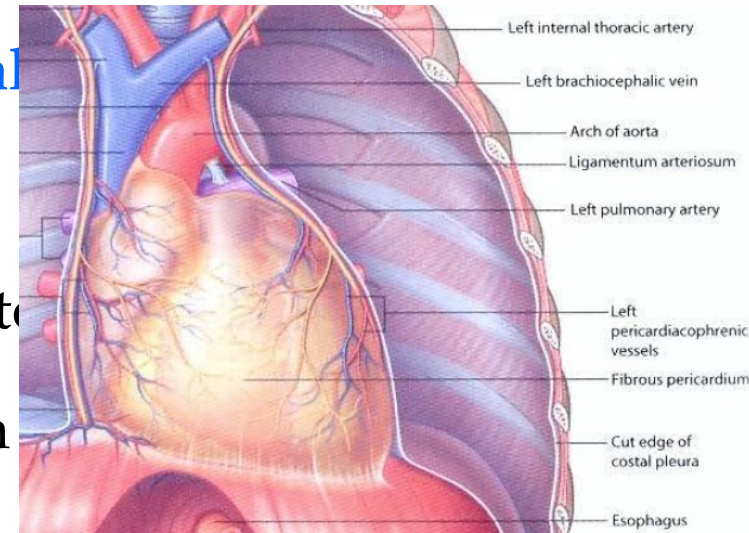
- ✓ outer -**fibrous pericardium**
- ✓ inner smooth- **serous pericardium**.
- A potential space b/n visceral & parietal serous pericardium is the **pericardial cavity**



Fibrous pericardium

It is attached

- A. **Anteriorly**- to the upper and lower end of body of sternum by **sternopericardial ligaments**
- B. **Posteriorly**- by loose connective tissue to structures in the posterior mediastinum
- C. **Superiorly**: directs upward & continuous with adventitia of the great vessels
- D. **Inferiorly**- to the central tendon of diaphragm



Function of fibrous pericardium

- Maintain the **central position** of the heart
- Keeps large vessels open
- Acts as a wall for the serous pericardium
- Prevents over loading & overdistention of the heart

Serous pericardium

- Double layered serous membrane
 - Outer layer – **Parietal** pericardium; fused with fibrous pericardium
 - Inner layer – **Visceral** pericardium (epicardium); attached to the heart and it extends onto the beginning of the **great vessels**, becoming **continuous** with the parietal layer
- The serous membranes, lubricated by **fluid** which is found in the **pericardial cavity** between the two layers, glide smoothly against one another during **heart activity**, creating a relatively **friction-free environment**.

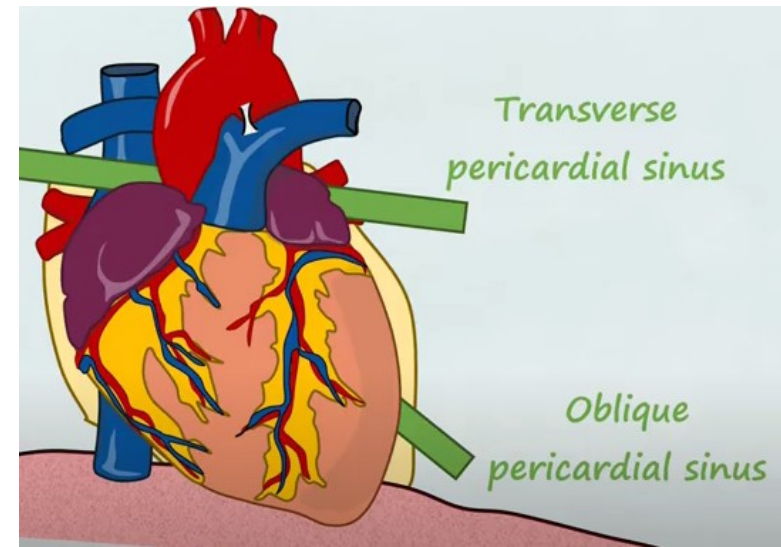
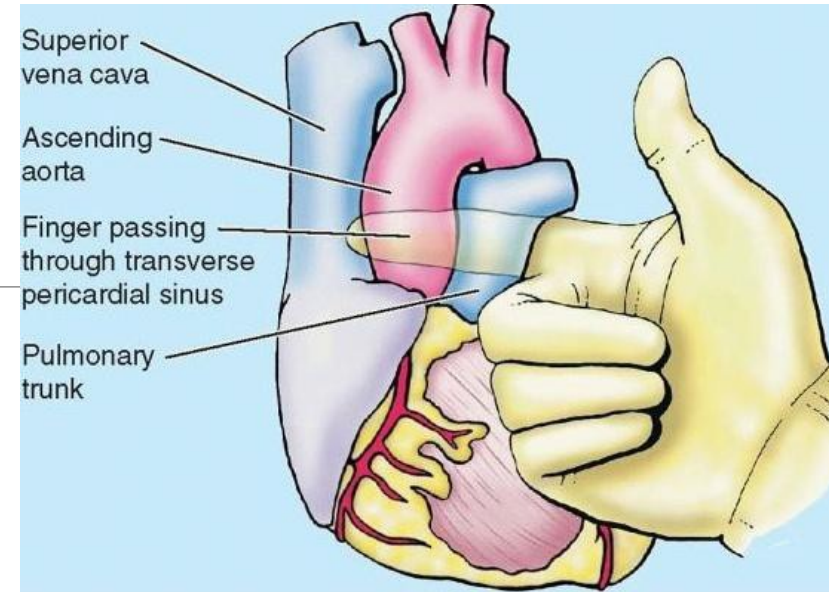
Pericardial Sinuses

- Formed by reflections of the parietal & visceral layers of the serous pericardium around the roots of great vessels

- Transverse Pericardial Sinus

- Oblique Pericardial Sinus

- **Importance** : By passing a ligature around these vessels, and then tightening the ligature, surgeons can stop or divert the circulation of blood in these arteries while performing cardiac surgery

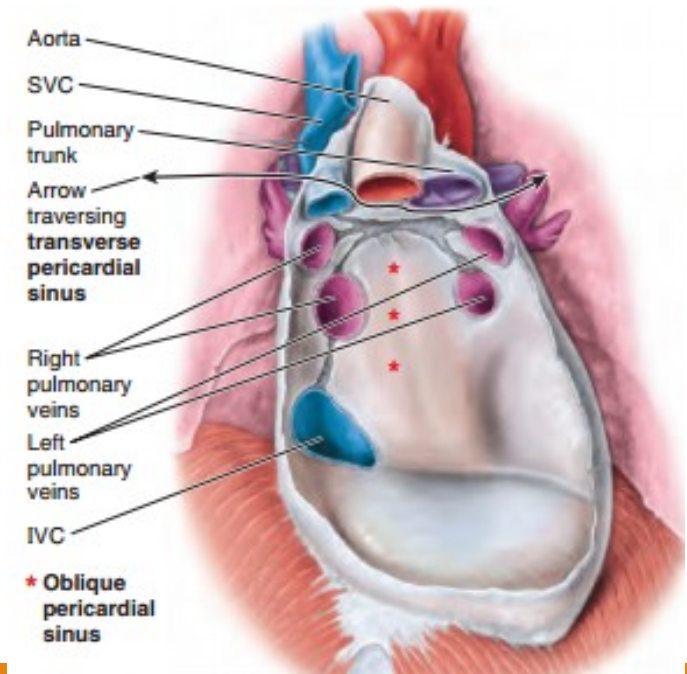
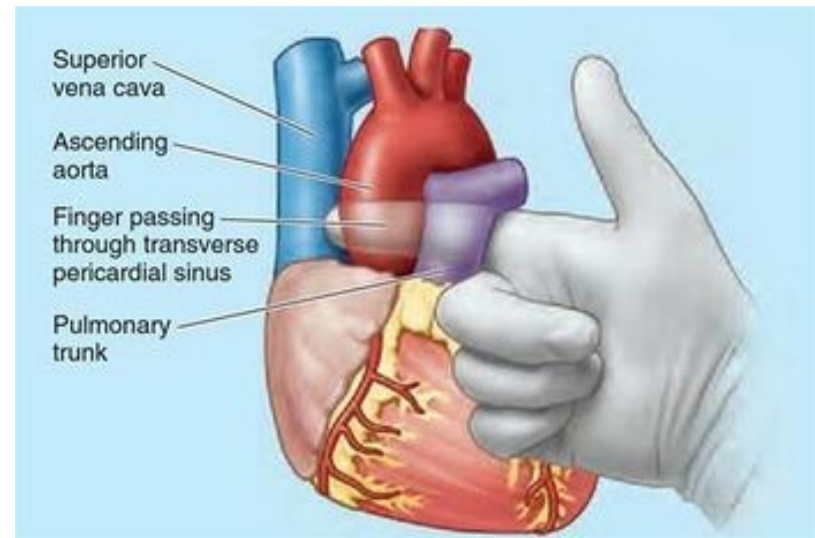


Transverse Pericardial sinus

- Is a transversely running passage within the pericardial cavity between **pulmonary trunk** and **ascending aorta** and the reflections of serous pericardium around them

Boundaries:

- **Ant:** Pulmonary trunk & ascending aorta.
- **Post:** SVC & Upper part of the 2 atria

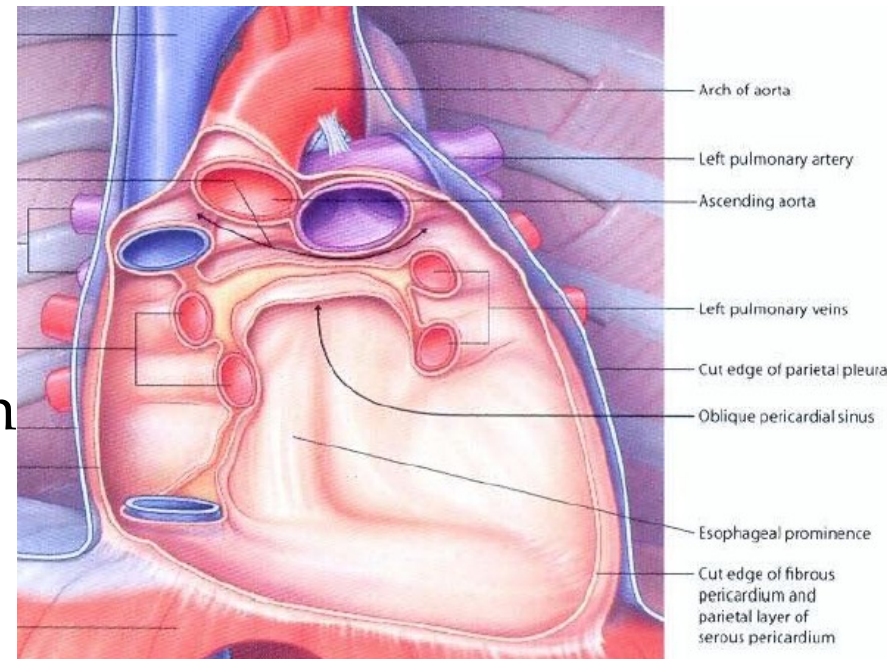


Oblique sinus

- a wide pocket-like recess in the pericardial cavity posterior to the base (posterior aspect) of the heart formed by the left atrium

Boundaries:

- **Laterally** -by the pericardial reflections surrounding the **pulmonary veins and IVC**
- **Posteriorly**- by the pericardium overlying the anterior aspect of the esophagus



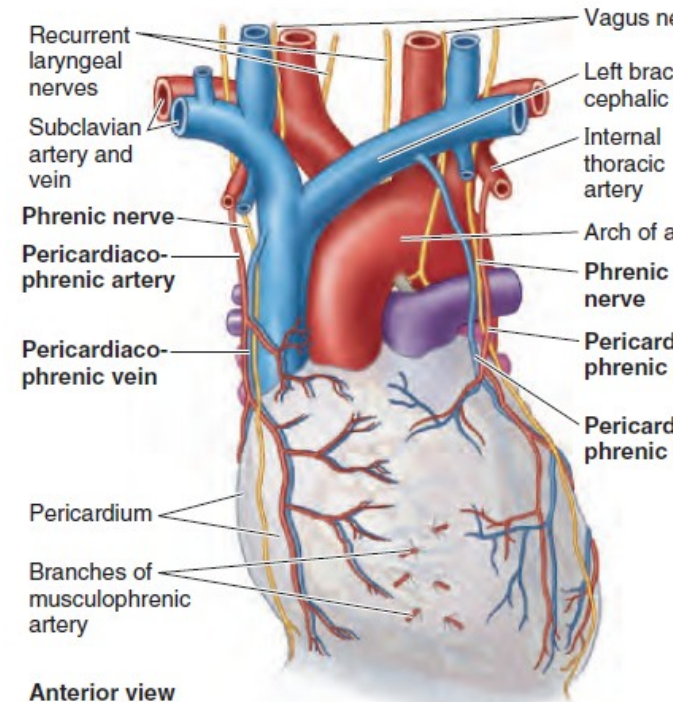
Blood supply and innervation of Pericardium

Fibrous and Parietal pericardium

- **Arteries** :pericardiacophrenic (main), musculophrenic, bronchial, esophageal, and superior phrenic arteries
- **Veins** :to pericardiacophrenic & azygos vv.
- **Nerve**: phrenic (sensitive to pain)

Visceral pericardium

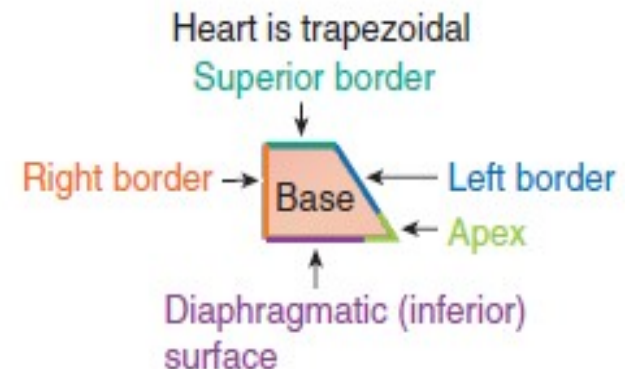
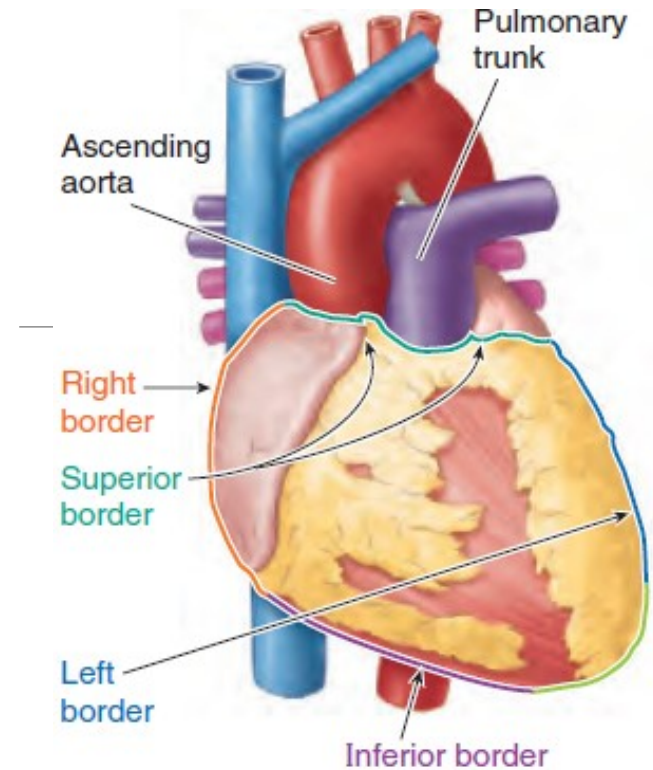
- **Artery** and **vein** : coronary
- **Nerve**: autonomic nerves of heart (not sensitive to pain)



HEART

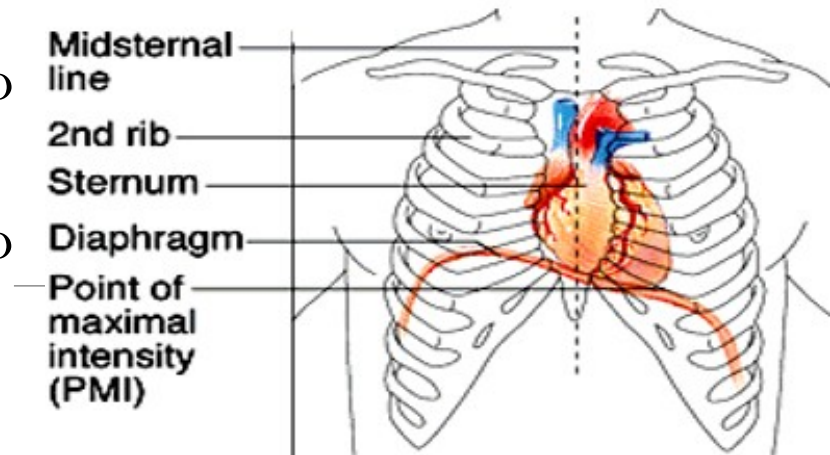
The heart is a hollow muscular organ

- Lies in the medial cavity of the thorax
 - Its $\frac{2}{3}$ rd lies to the **left** of the midsternal line
 - It is pyramidal in shape with an **apex**, **base**, **four surfaces** and with **four borders**
- Its right side receives **oxygen poor blood** from the body and then pumps it to the lungs
 - Its left side receives **oxygenated blood** from the lungs and then pumps it to the body



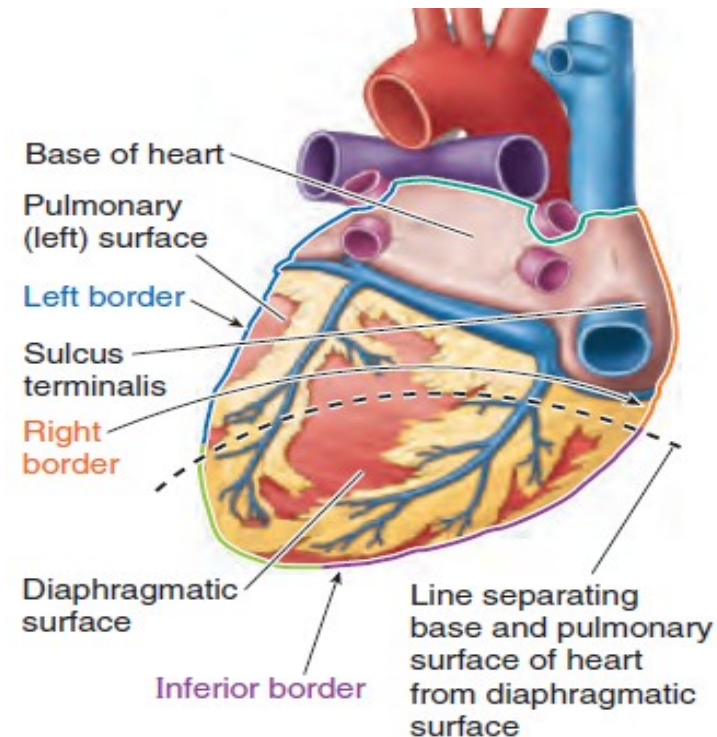
Apex

- points downwards, forwards and to the left
- lies in **left 5th ICS** just medial to MCL
- formed by mainly by **left ventricle**



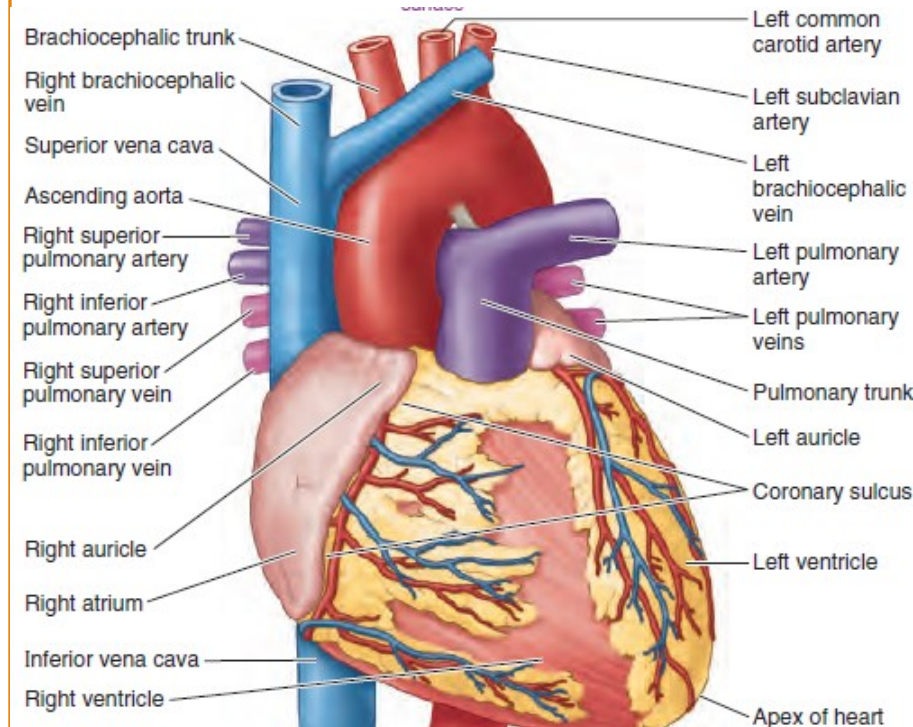
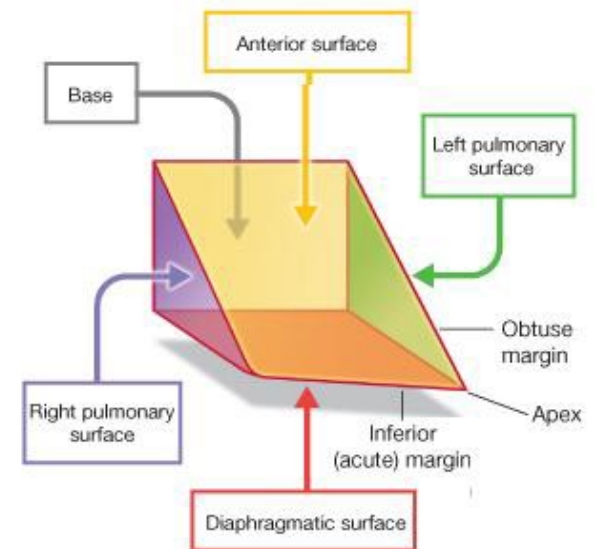
Base : directs to right shoulder

- Between bifurcation of **pulmonary trunk** and **coronary groove**
- Formed mainly by **LA** and small part of **RA**
- Faces posteriorly toward the bodies of vertebral **T6 –T9**
- Receives the **pulmonary vein**, **superior** and **inferior venae cavae**



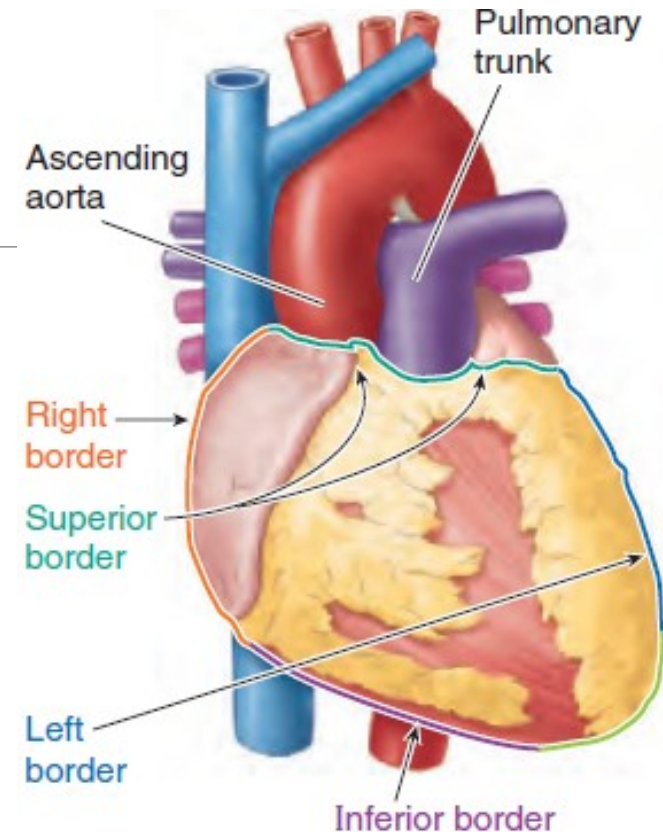
Surfaces of the heart

- **Sternocostal (anterior) surface:**
 - ✓ Faces anteriorly
 - ✓ Formed mainly RV
- **Diaphragmatic (inferior) surface:**
 - ✓ Directed downwards
 - ✓ formed by mainly by LV & partly by RV
- **Right pulmonary surface:**
 - Formed by **RA**
- **Left pulmonary surface:**
 - ✓ Formed by **LV**
 - ✓ It forms the **cardiac impression** in the left lung



Border

- **Right** - formed by **RA**, and extends between SVC and IVC
- **Inferior** –nearly horizontal, formed mainly by **RV**, slightly by **LV**
- **Left** – oblique, nearly vertical, formed mainly by **LV** and partly by **left auricle**
- **Superior** – slightly oblique, formed by **two atria** and **auricle**
 - the **ascending aorta** and **pulmonary trunk** emerge from this border



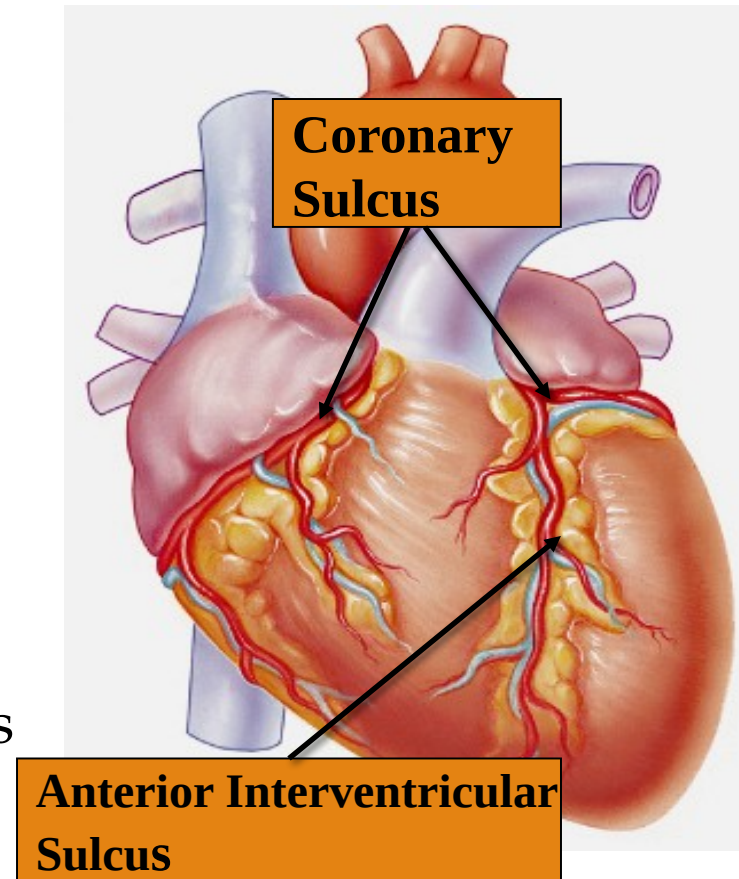
Sulci of the heart

➤ The coronary sulcus: (atrioventricular groove)

- ✓ Circles the heart, separating the atria from the ventricles

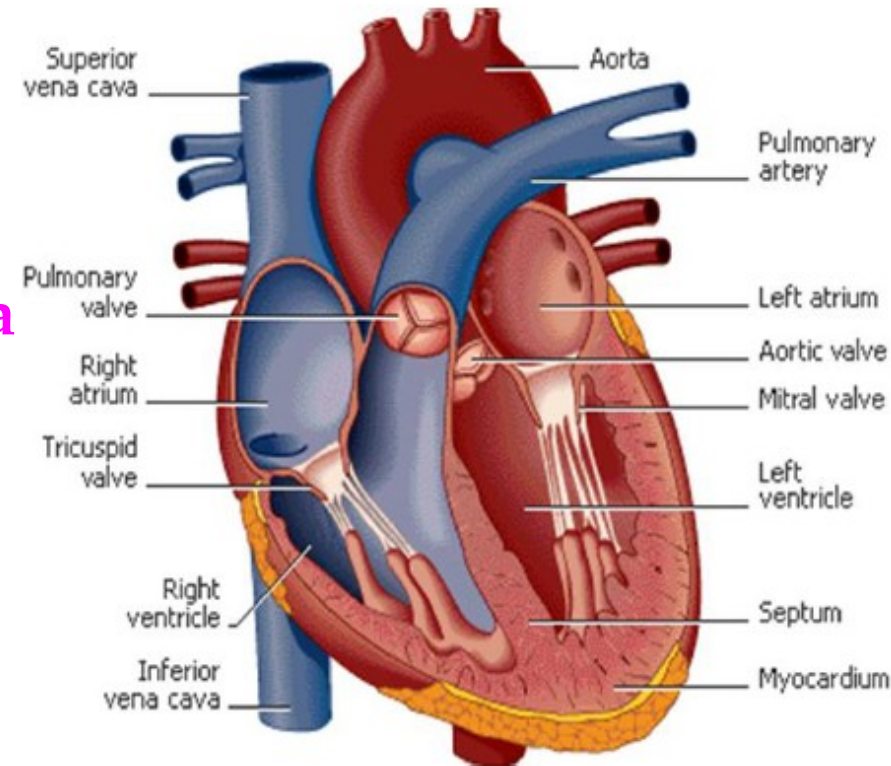
➤ The interventricular sulci (anterior and posterior):

- ✓ Extend vertically through ventricles both anteriorly and posteriorly
- ✓ Separate the ventricles externally into **right** and **left** ventricles



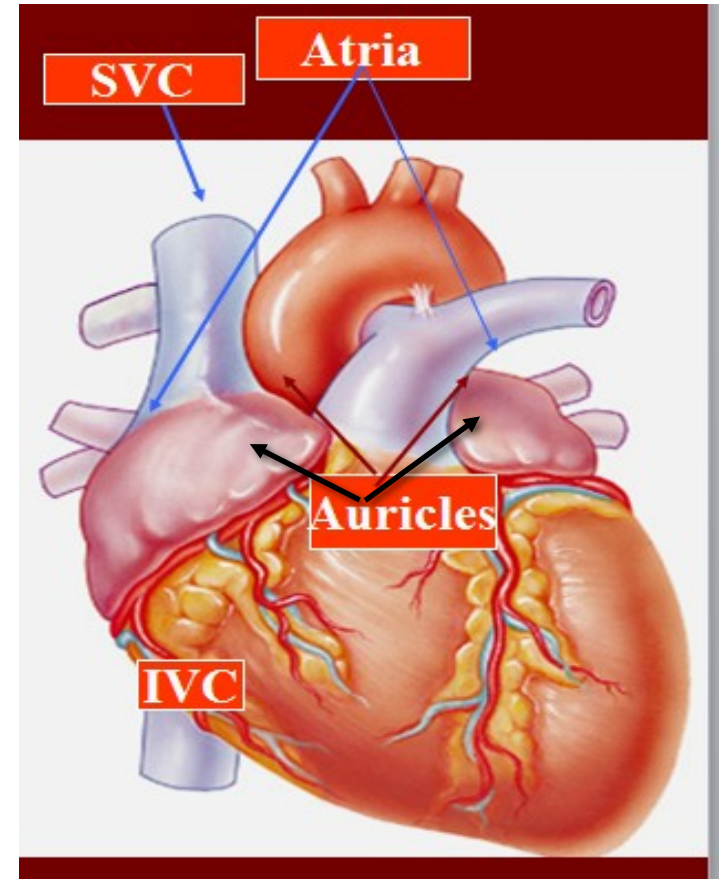
CHAMBERS OF THE HEART

- The heart is divided into **four chambers**:
- The **interatrial septum** separates the **right** and **left atria**
- The **interventricular septum** separates the **right** and **left ventricle**
- Upper part of **each atrium** has an appendage called **auricle**



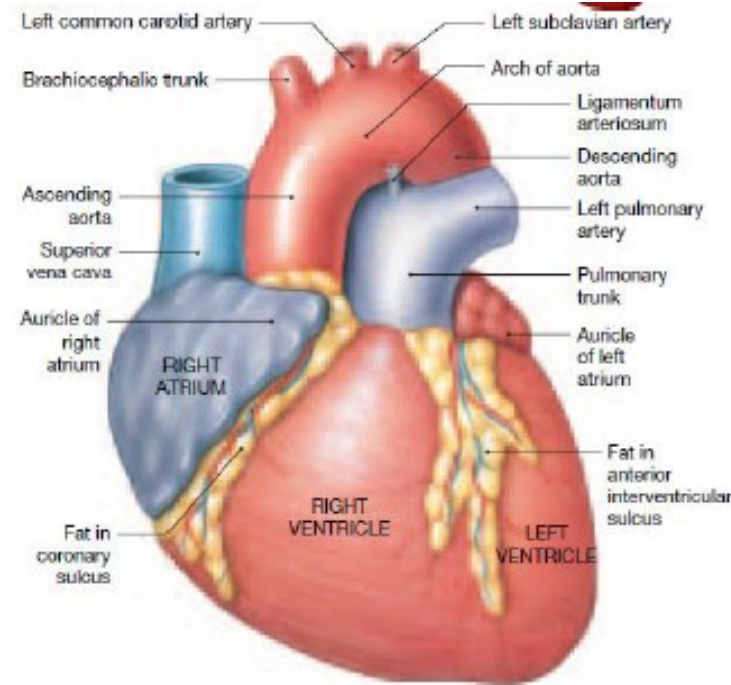
Right atrium

- Receives venous blood from the body through **SVC**, **IVC** and **CS**
- pumps to RV through right **AV orifice**
- Forms the **right border**, the **sternocostal surface** and **base** of heart



Right atrium: external features

- Elongated vertically
- Receiving SVC at upper end and IVC at lower end
- Upper end is prolonged to left to form **right auricles**
- **Sulcus terminalis** - a shallow **groove** which separates smooth and rough parts of the atrial wall **externally** along right border run from **SVC** to **IVC** vertically
 - produced by internal muscular elevation (**cristae terminalis**)
 - upper end is **land mark** of SA node



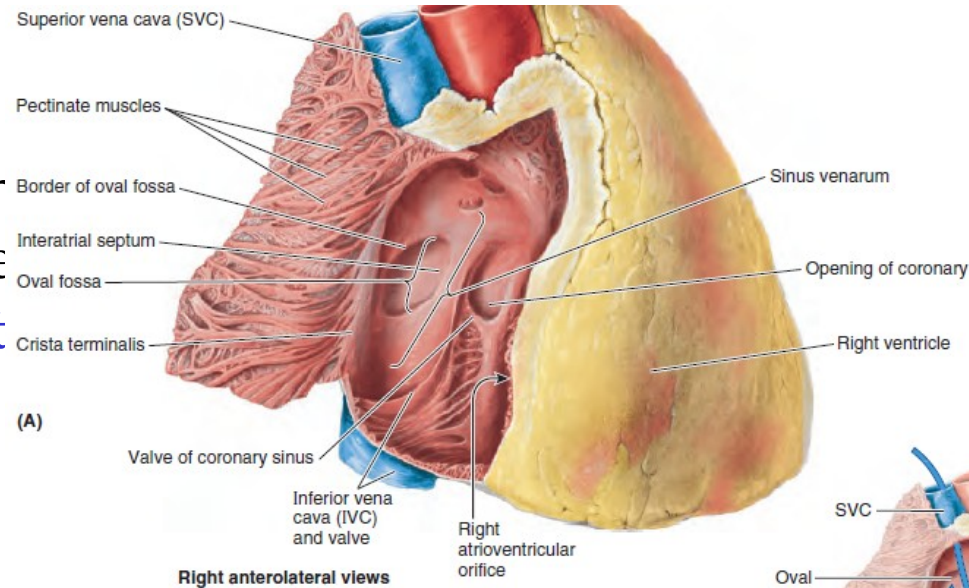
Right atrium: internal features

□ The smooth posterior part (sinus venarum)

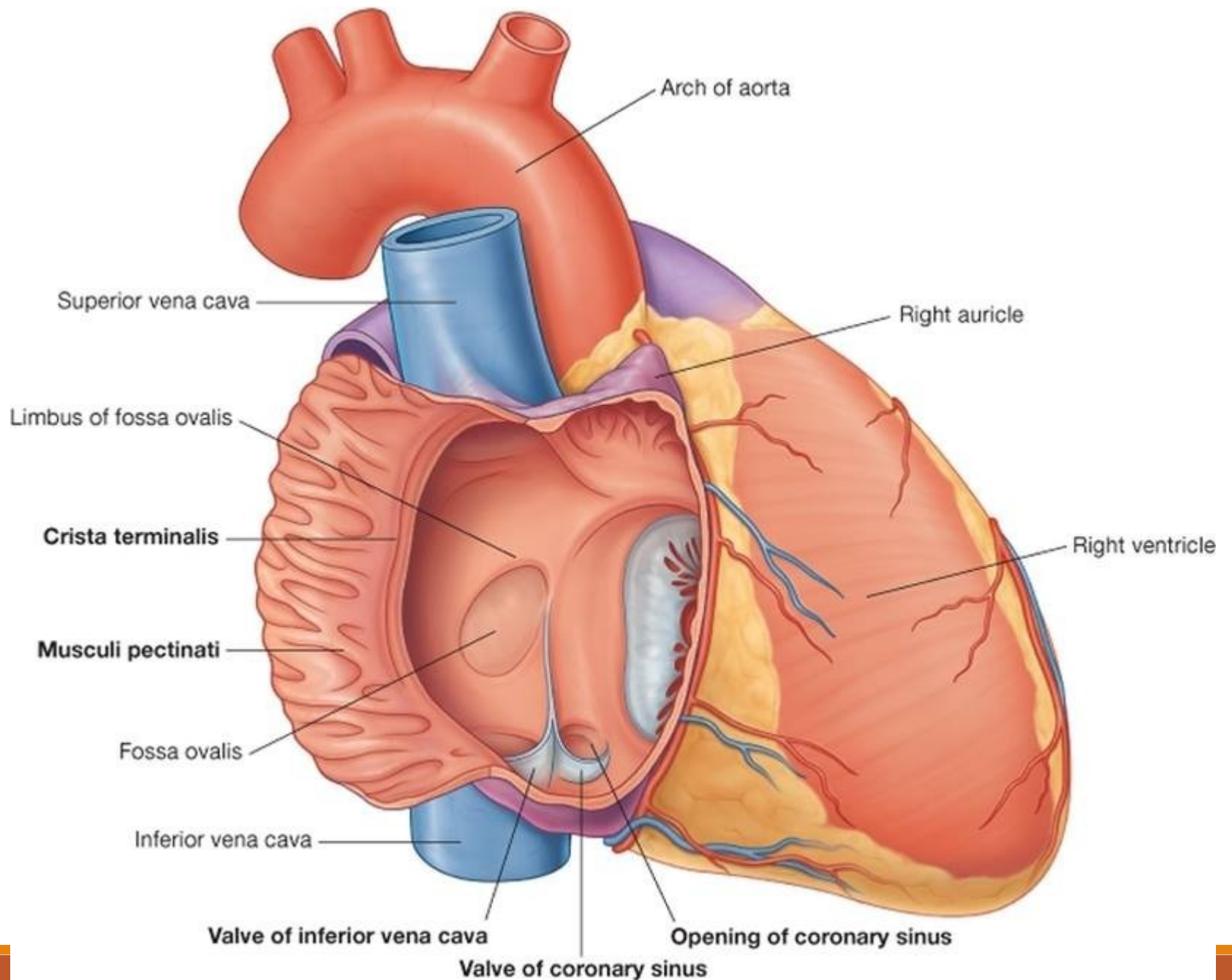
- ✓ Derived from embryonic **sinus venosus**
- ✓ SVC & IVC opens into it
- ✓ **Coronary sinus** opens between IVC orifice and right AV orifice

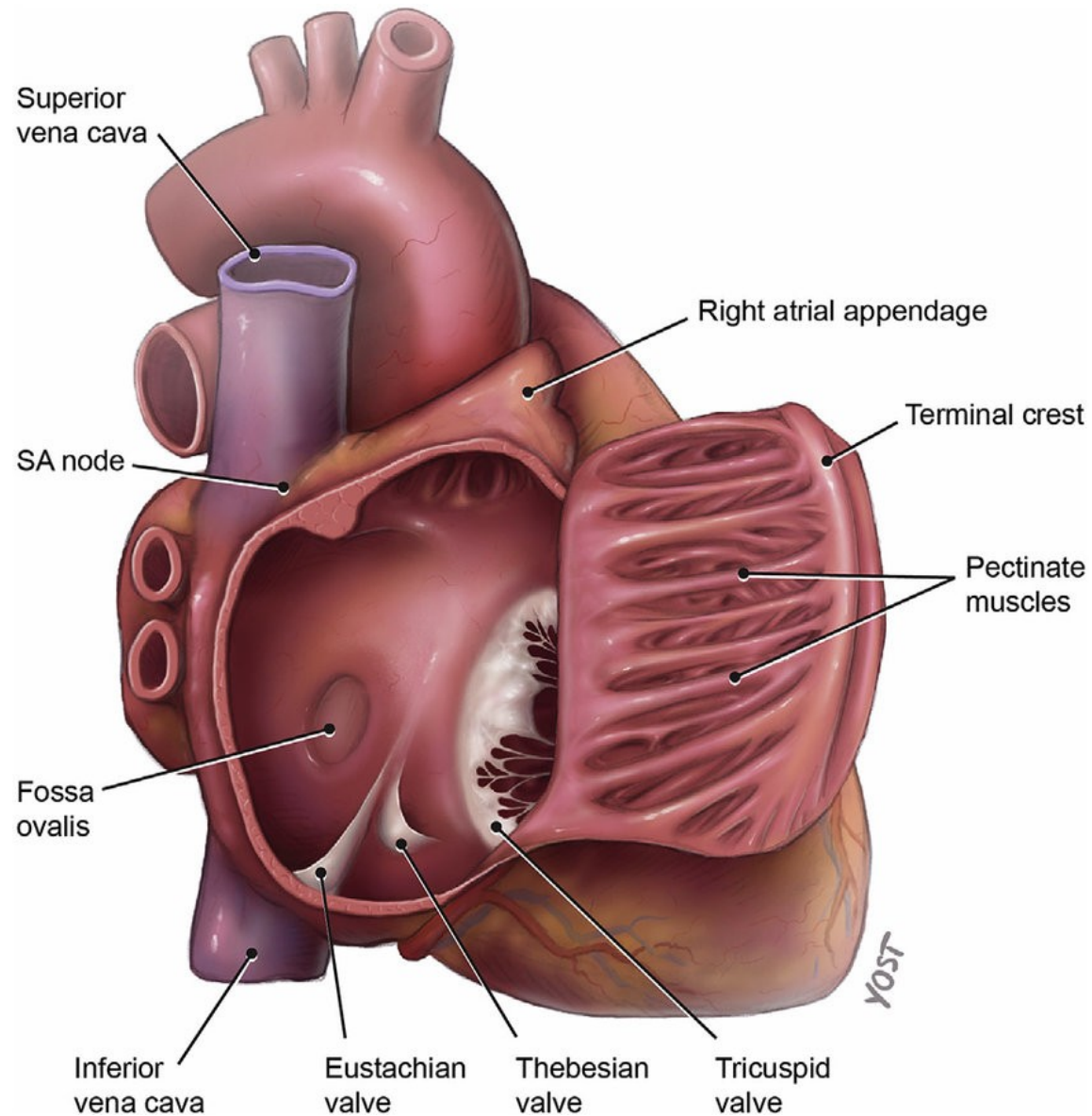
□ The rough anterior part (pectinate part)

- ✓ Present a series of transverse muscular ridges (**pectinate muscle**)
- ✓ Arise from **crista terminalis** and run towards **AV orifice**



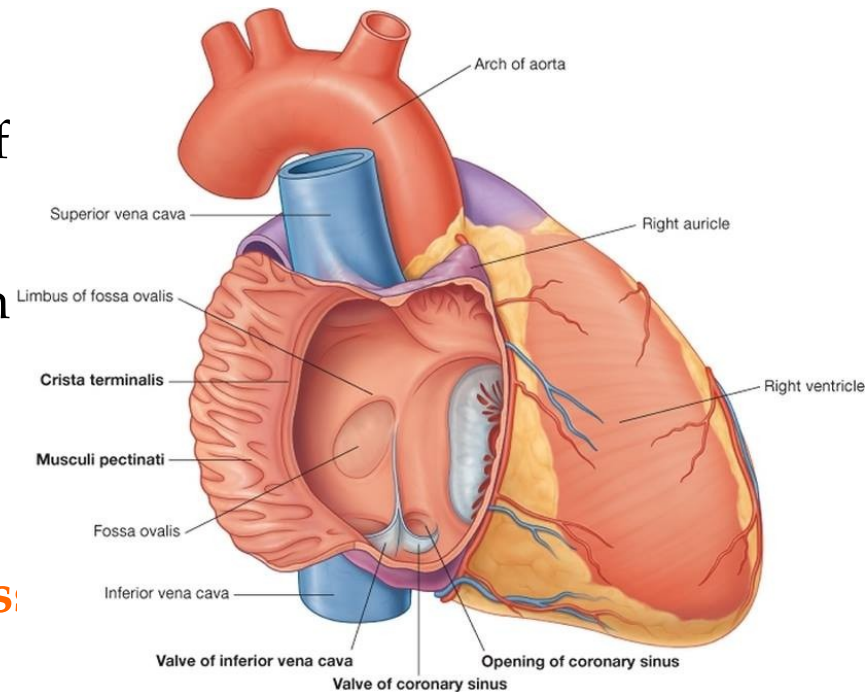
Crista terminalis, is a smooth, muscular ridge begins on the opening of the **SVC** & extends down to anterior lip of **IVC**





Structures in the Right atrium

1. **Pectinate muscles**, muscular ridges, which fan out from the **crista terminalis**
2. **opening of coronary sinus**, opens medial to **opening of IVC**.
3. **valve of CS & IVC** (Associated with opening of coronary sinus, & opening of IVC are **small folds of tissue**).
3. **interatrial septum**: separating Rt atrium from Lt atrium.
3. **fossa ovalis** : a depression in **interatrial septum**
 - its has prominent margin, the **limbus fossa ovalis** (border of oval fossa).
6. **openings of smallest cardiac veins** are scattered along walls of Rt atrium

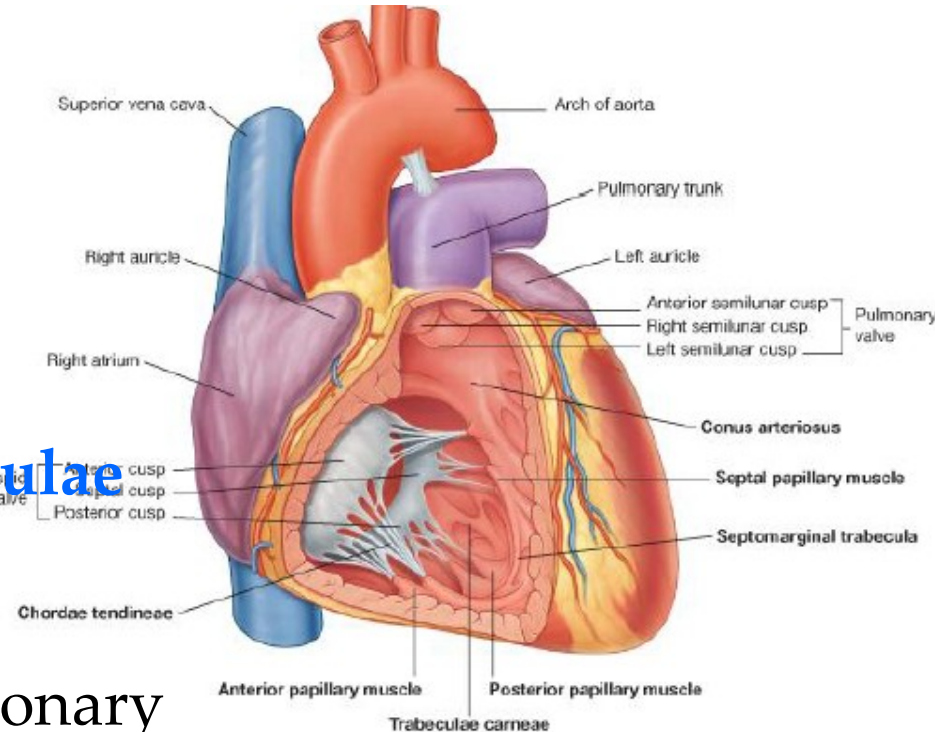


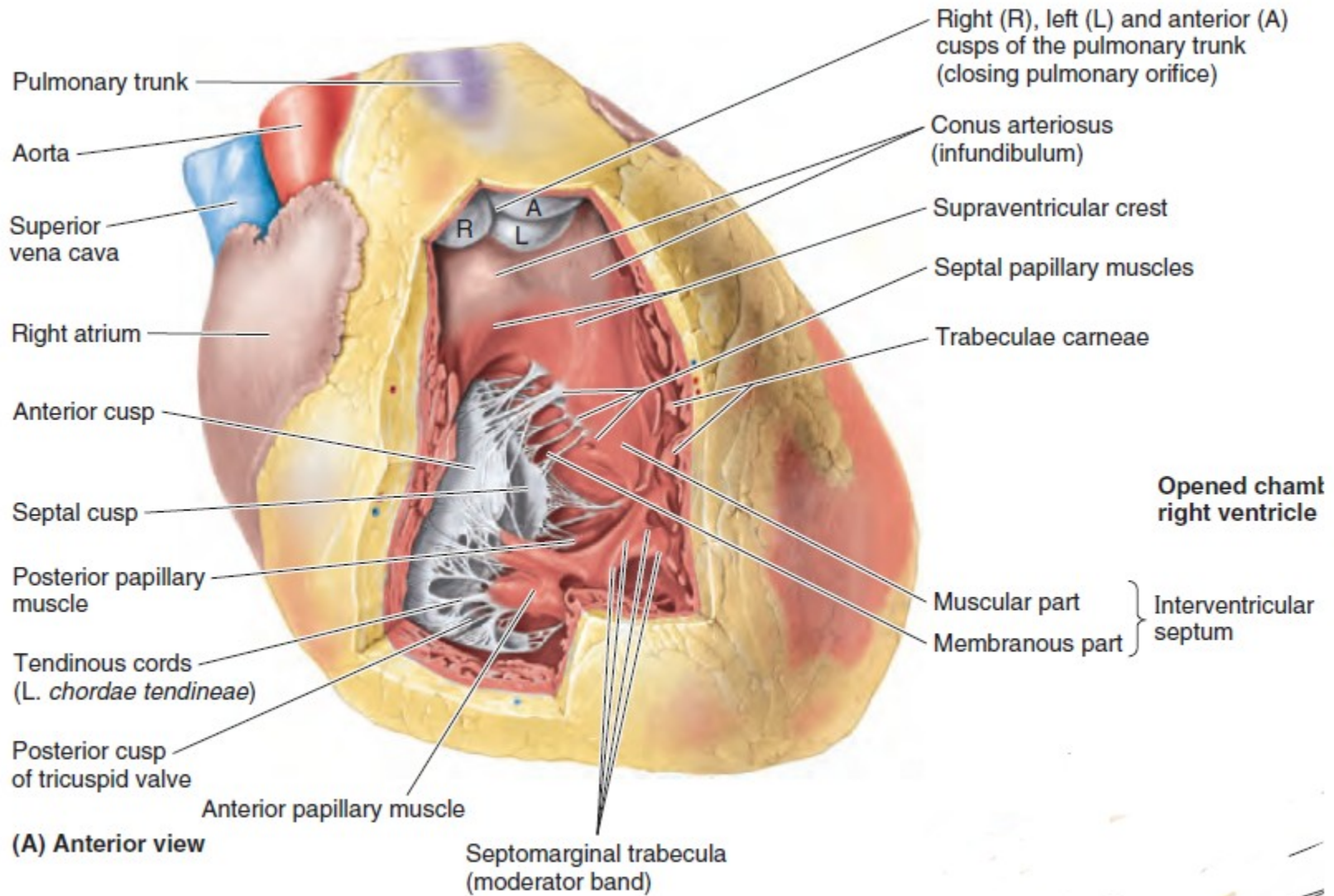
Right ventricle

- forms most of the **anterior surface**, small part of **diaphragmatic surface** and almost **entire inferior border**
- Triangular in shape

Internal structure

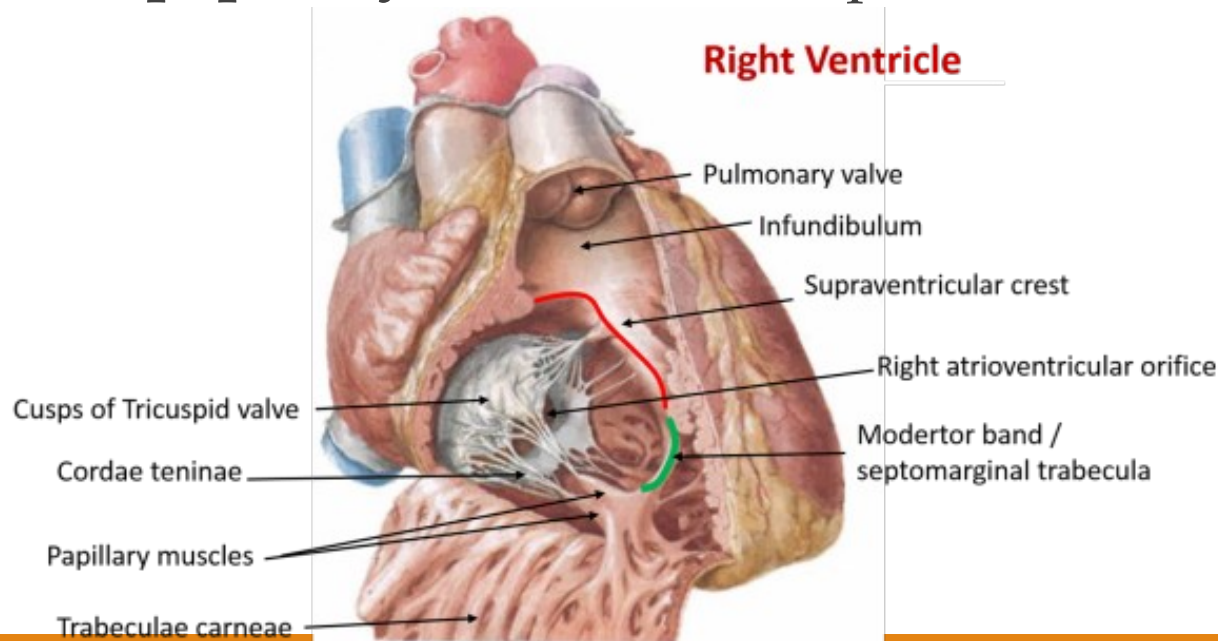
- **Inflow part** – **rough** due to muscular ridges called **trabeculae carneae**
- **Outflow part** – **smooth**, pulmonary trunk arise
 - separated from inflow part by **supraventricular crest**





Right ventricle: Internal structure ...

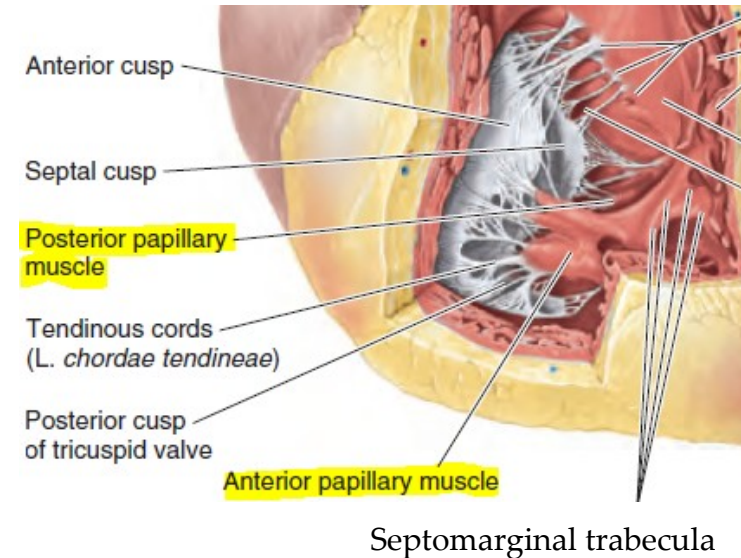
- ✓ **Conus arteriosus (infundibulum):** located at the outflow tract of RV, which leads to the **pulmonary trunk**.
- ✓ **Trabeculae carneae:** are numerous **irregular muscular** structures on the walls of RV.
- ✓ **Chordae tendineae** : are tendon-like fibrous cords, that connect the **papillary muscles** to cusps of the **tricuspid valve**.



Right ventricle: Internal- Structures...

✓ **papillary muscles** are conical muscular projections with bases attached to the **ventricular wall**

- There are three papillary muscles in the right ventricle
 - ✓ **Anterior papillary muscle:** arises from the **anterior wall** of the right ventricle;
 - ✓ **Posterior papillary muscle:** arising directly from the **posterior ventricular wall**
 - ✓ **Septal papillary muscle:** emerging directly from the septal wall.

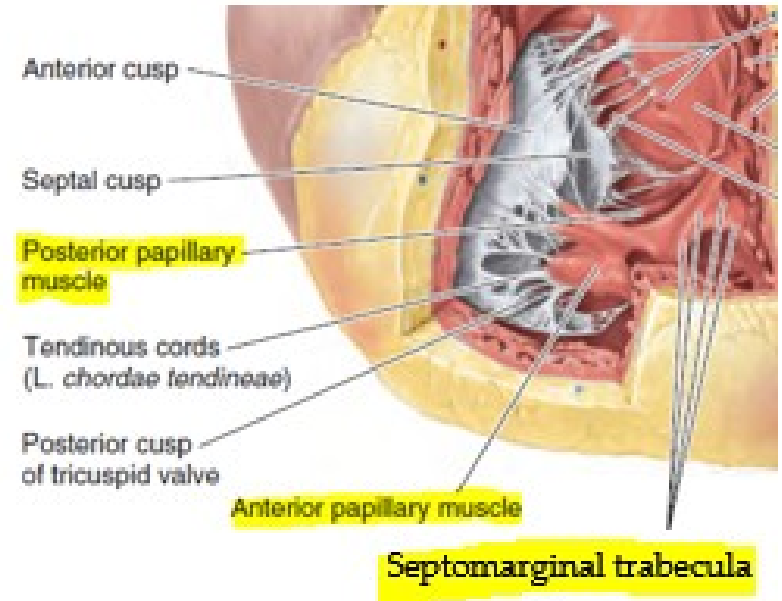


Right ventricle: Internal- Structures...

✓ Septomarginal trabecula

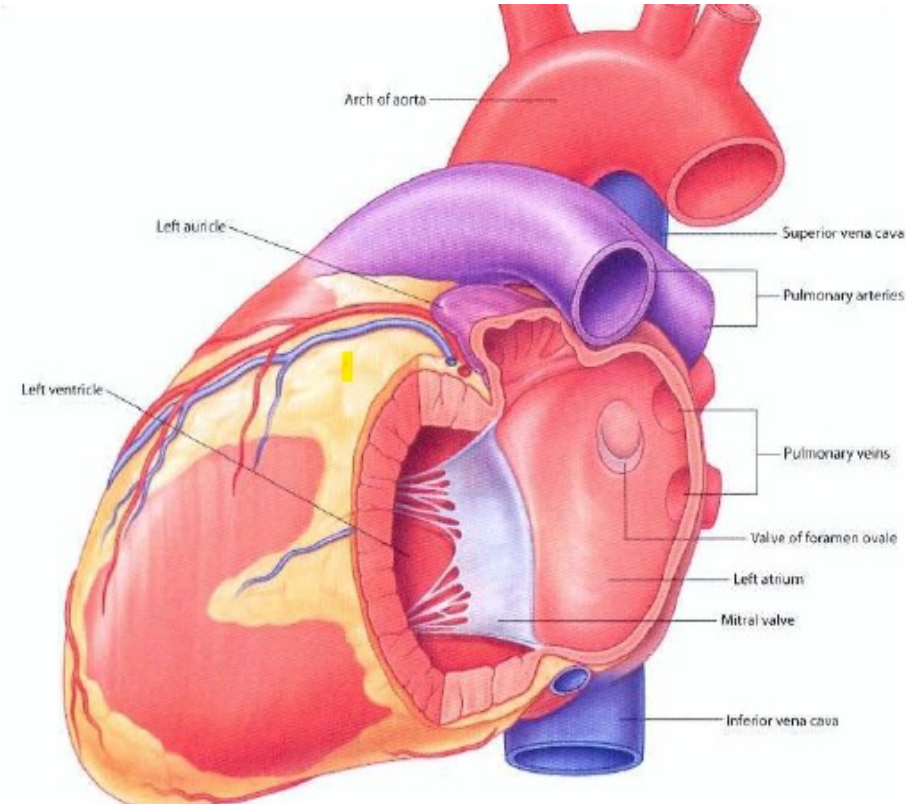
(moderator band), forms a bridge b/n lower part of IVS & base of anterior papillary muscle.

■ This trabecula is important because it carries part of the **right branch of the AV bundle**, to the **anterior wall of rt ventricle**.



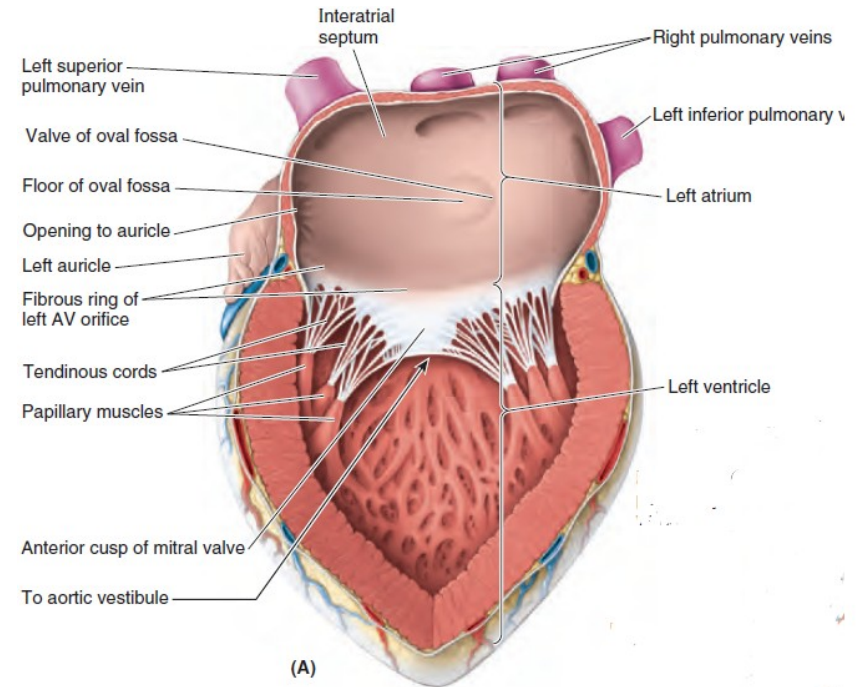
Left atrium

- Receives oxygenated blood from **PV**
- Pumps through the left **AV** orifice into the **left ventricle**
- Communicates with LV thru the **mitral valve (bicuspid valve)**.
- Its posterior half presents openings of the **four pulmonary veins**
- The **interatrial septum** is part of the anterior wall of Lt atrium which separates Rt atrium Lt atrium



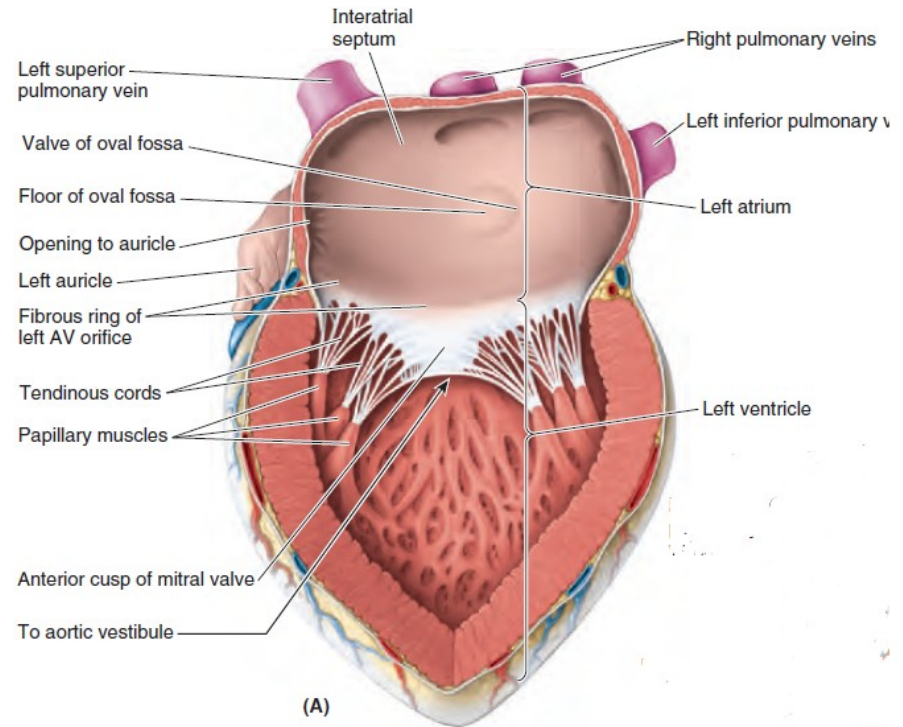
Left atrium...

- *Semilunar depression* in the interatrial septum indicates the floor of the **oval fossa**
- A left **AV orifice** through which the left atrium discharges the oxygenated blood from the **pulmonary veins** into the **left ventricle**



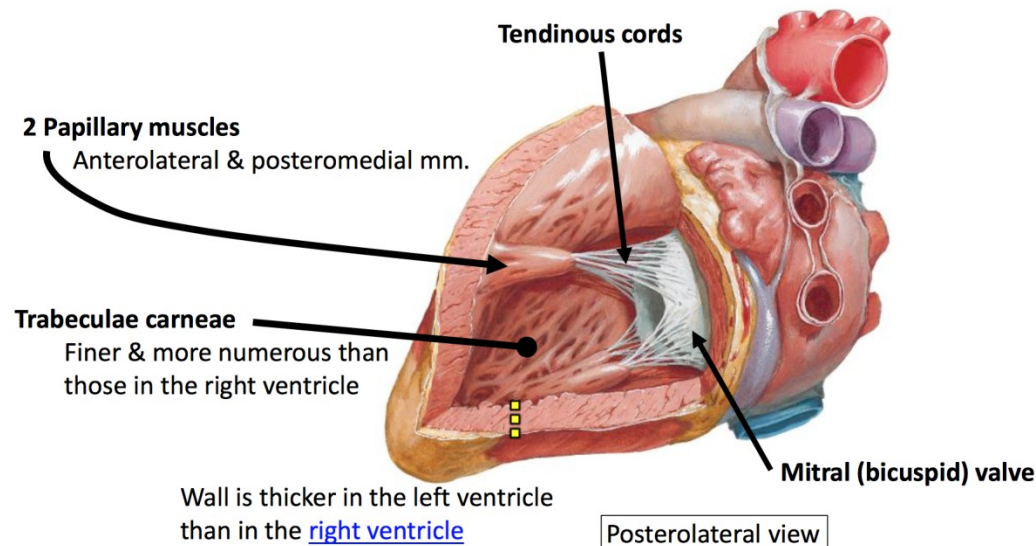
Left Ventricle

- forms the **apex** of the heart, most of **diaphragmatic** surface and **left border**
- The walls of the left ventricle are 2-3 times **thicker** than that of the **right** ventricle due to the work load in Lf ventricle



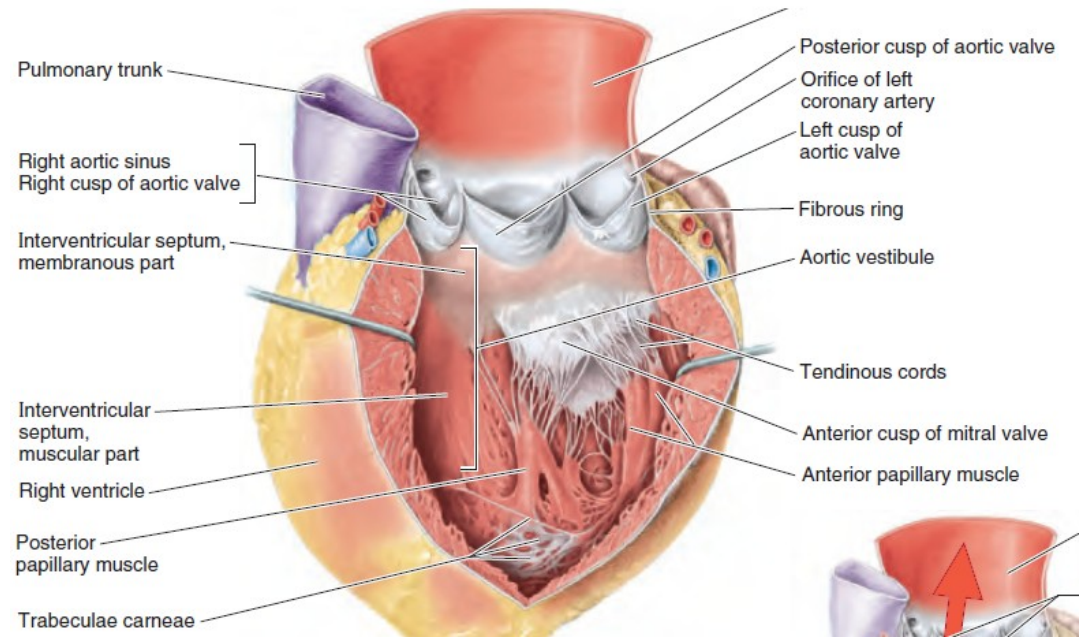
Left Ventricle

- Blood enters ventricle thru the **left AV orifice**.
- **trabeculae carneae** are finer and more numerous than those of the right ventricle.
- Two papillary muscles, **anterior & posterior papillary muscles** which are larger than those of the Rt ventricle
- Valve cusps connect to the **papillary** muscles by **chordae tendineae**



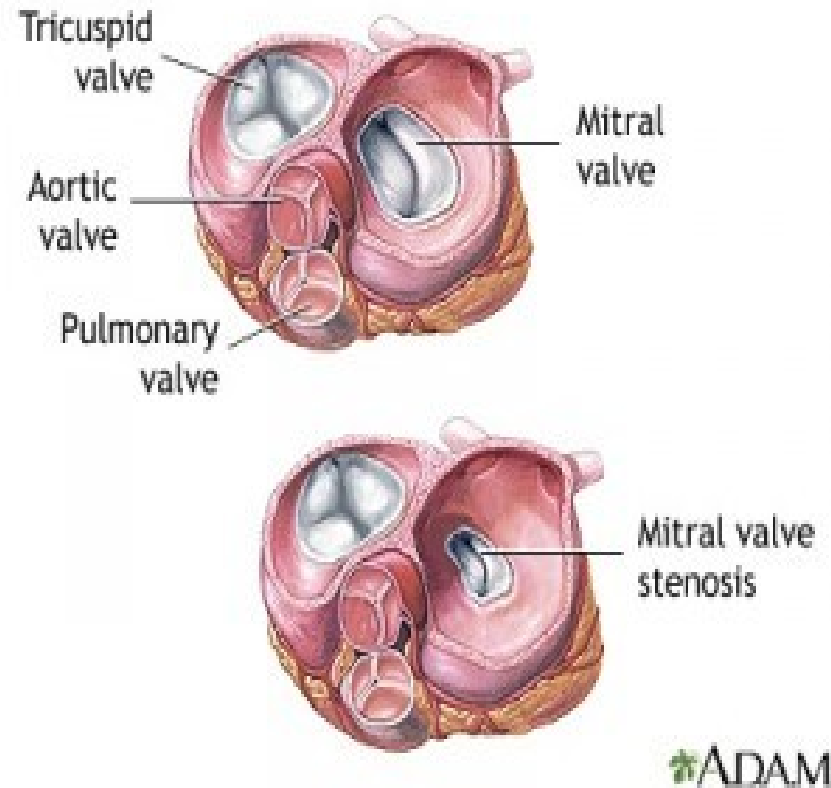
Left Ventricle

- In the anatomic position, Lt ventricle is somewhat **posterior** to Rt ventricle
- The smooth superior part is referred to as **aortic vestibule**, leads to aortic orifice



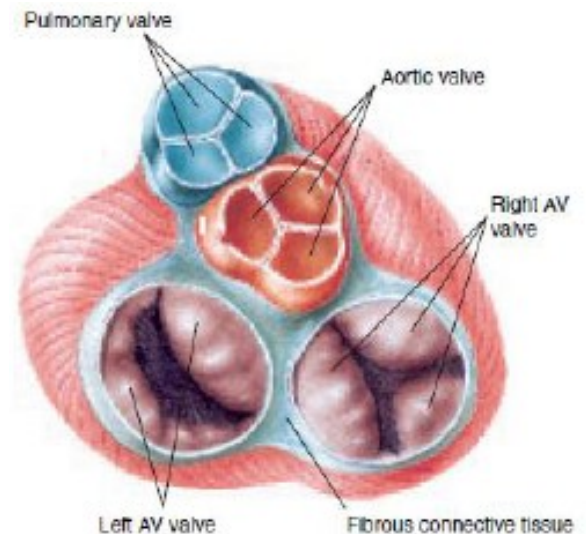
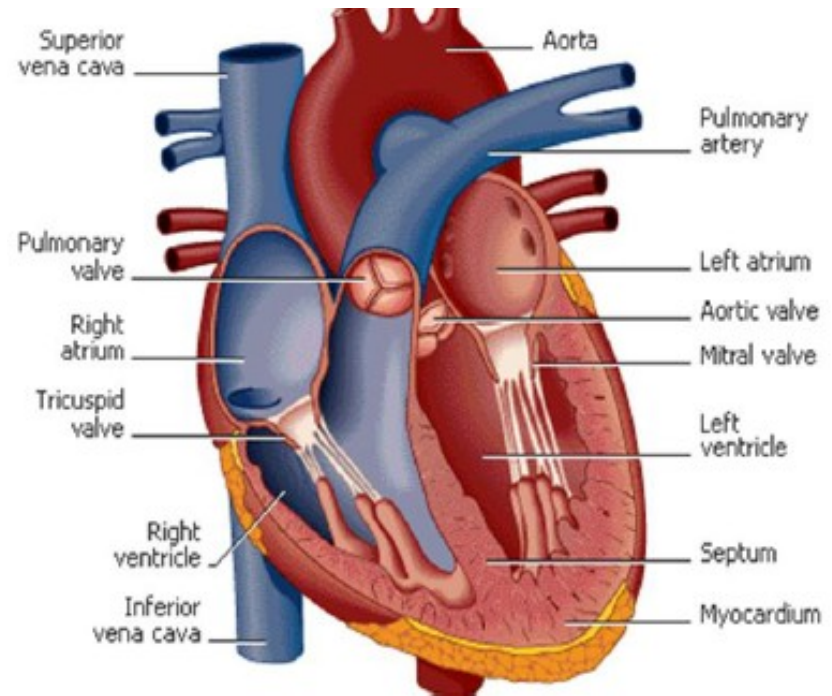
Heart valves

- **prevent back flow** of blood and ensure that blood flows in the proper direction through the heart.
- The valves function is to maintain **blood flow in one direction**.
- Valves open and close in response to changes in **pressure**



Four heart valves are

- **AV valves**
 - **Tricuspid valves** - the right AV valves
 - **Bicuspid valves** - the left AV valves
- **Semilunar valves**
 - **Aortic valve** - Left Ventricle and Aorta
 - **Pulmonary valve**- Rt Ventricle and Pulmonary Trunk



Atrioventricular valves

- located at each atrio-ventricular junction

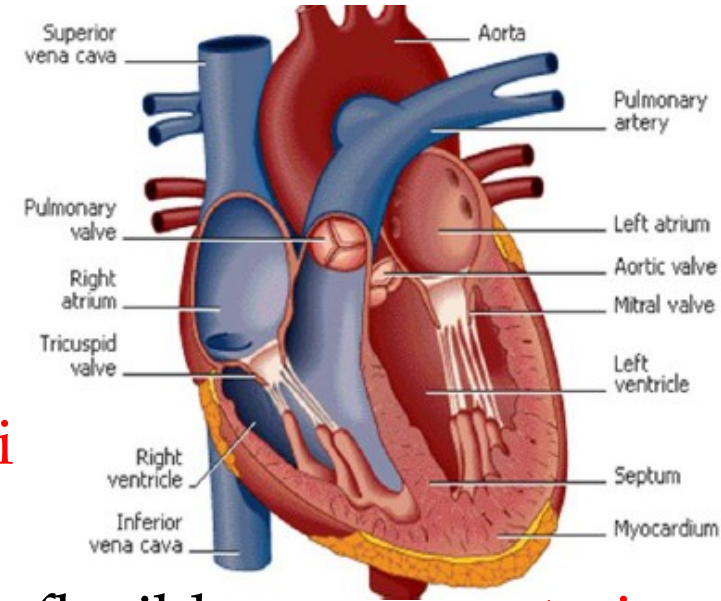
- The **right AV valve (tricuspid)** has three flexible cusps - **anterior, posterior**

& septal

- The left AV valve (bicuspid)** has two flexible cusps - **anterior and posterior**

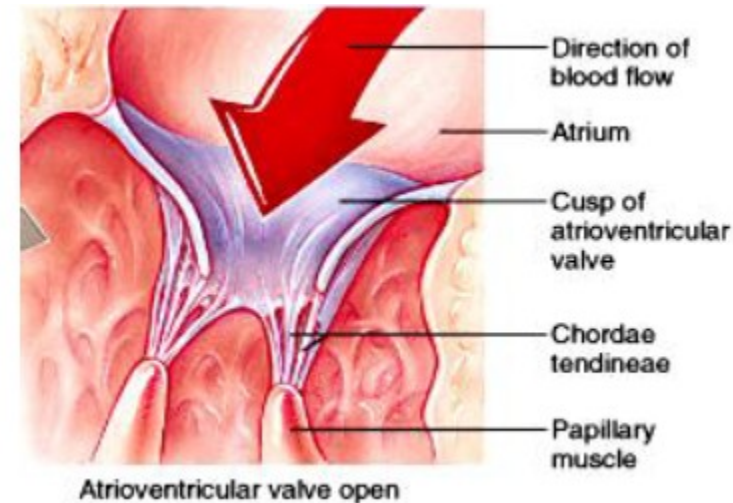
- The mitral valve is located posterior to the sternum at the level of the 4th costal cartilage.

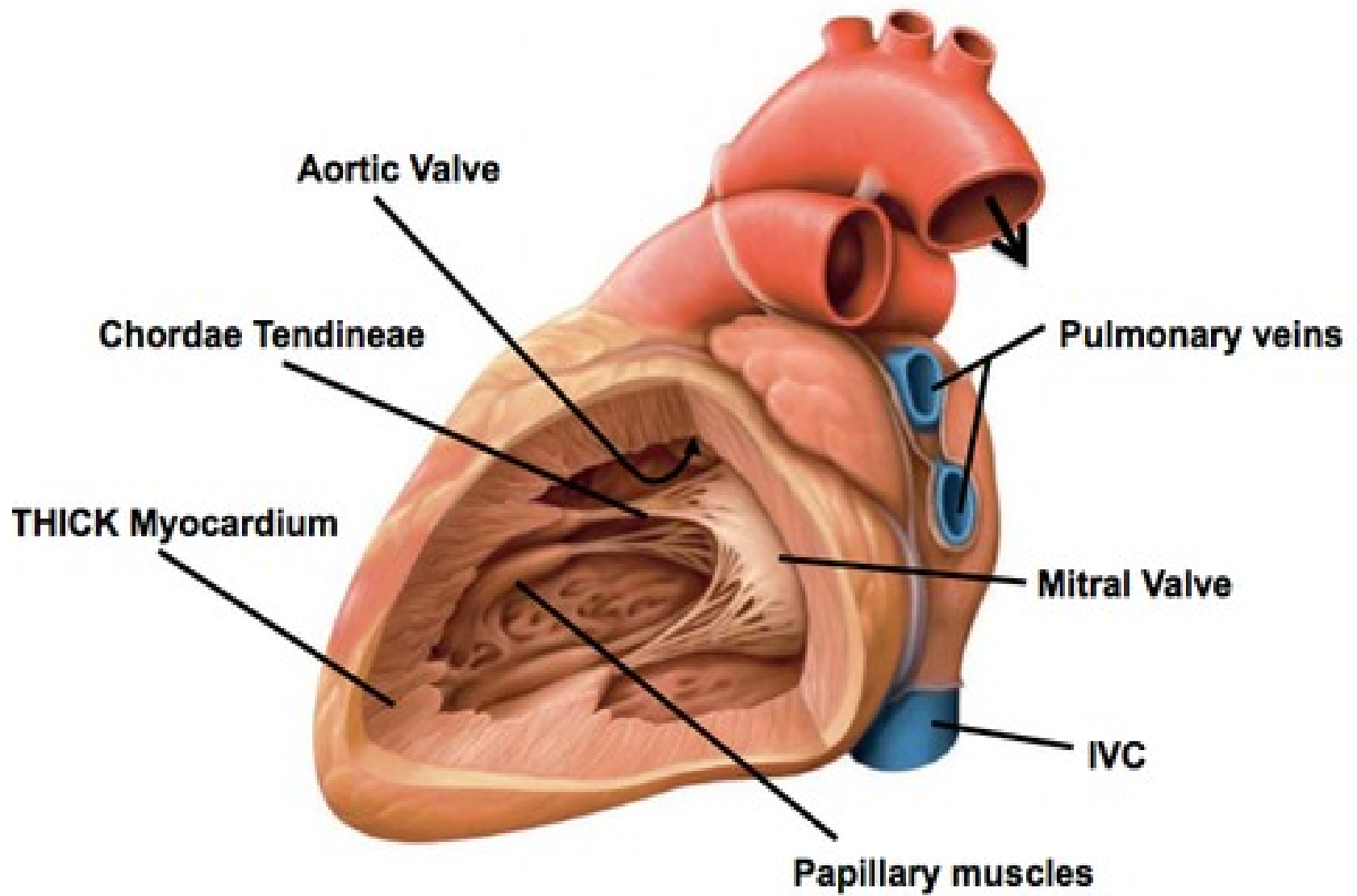
- Each cusp is attached to the corresponding papillary muscle by **chordae tendinae**.



Components of the AV valves

- **Cusps** – projection of the endocardium
 - has attached (to fibrous ring) and free border
 - two surfaces : atrial – smooth and ventricular – rough
- **Chordae tendineae**
 - connect the free ventricular ends of **cusps** with the apex of **papillary muscles**
 - Prevents separation and inversion of the cusps during systole
- **Papillary muscles**: projection of the myocardium of the ventricle





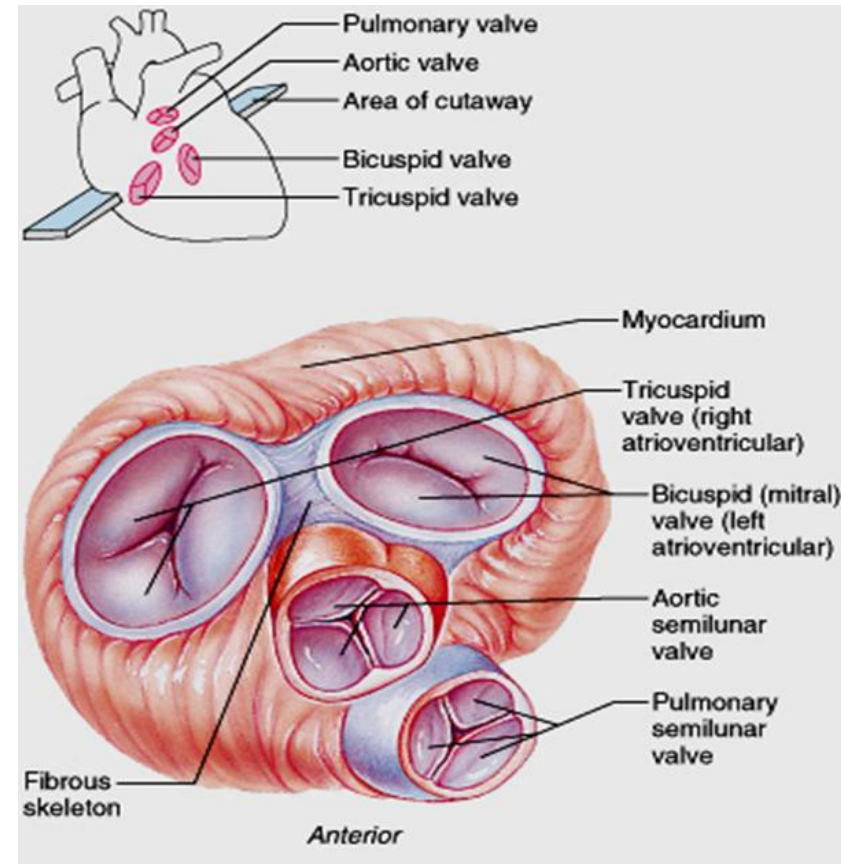
Semilunar Valves

➤ Aortic valve:

- ✓ lies at the base of ascending aorta
- ✓ Which is made up of 3 pocket like cusps:-**posterior, right & left**

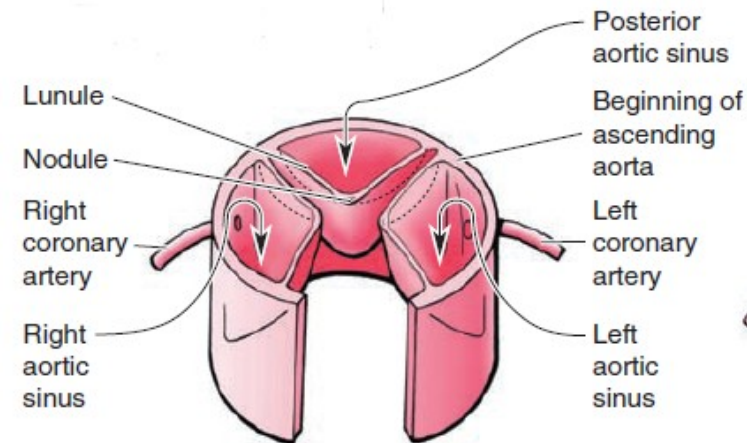
➤ Pulmonary valve

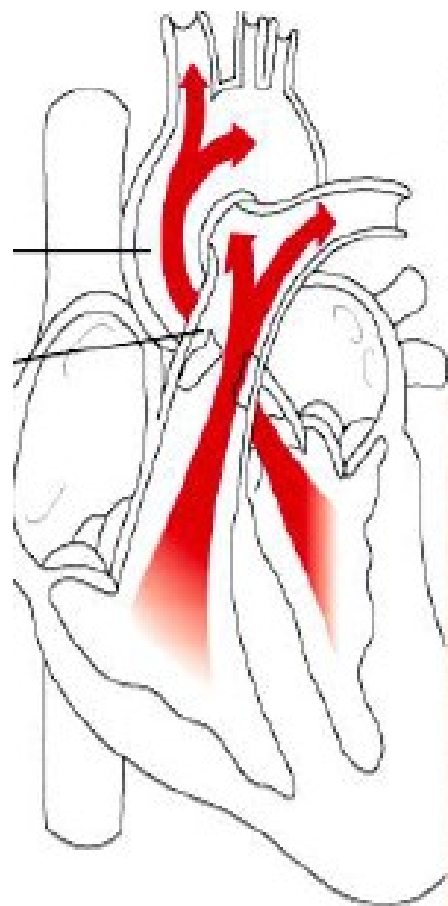
- ✓ lies at the base of pulmonary trunk
- ✓ Which is made up of 3 pocket like cusps:-**anterior, right & left**



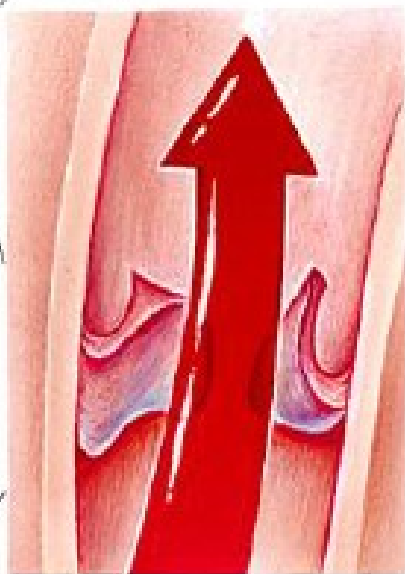
Semilunar Valves...

- Each cusp is **concave superiorly**
- Semilunar cusps **do not have tendinous cords** to support them.
- the spaces between dilated wall of the vessel and on each cusp of the semilunar valves is called as **pulmonary/aortic sinuse**
- Blood in the sinuses prevents cusps from sticking to wall of the vessel
- The mouth of the **right & left coronary artery** are in the **right & left aortic sinus** respectively ,and no artery arises from the **posterior aortic (non-coronary) sinus**

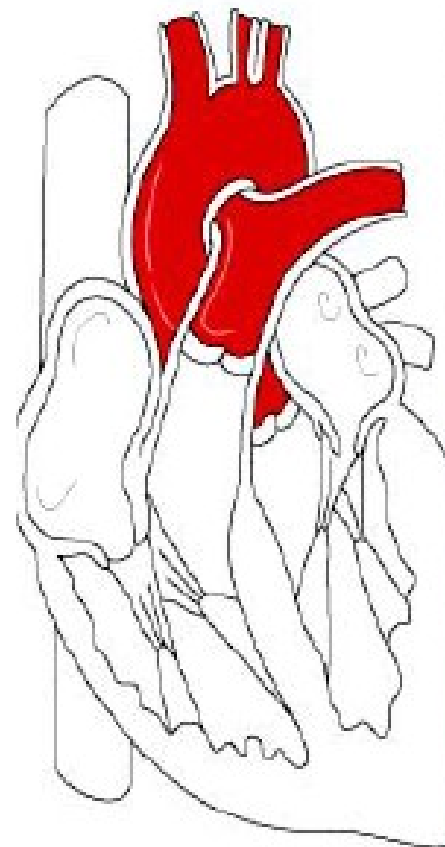




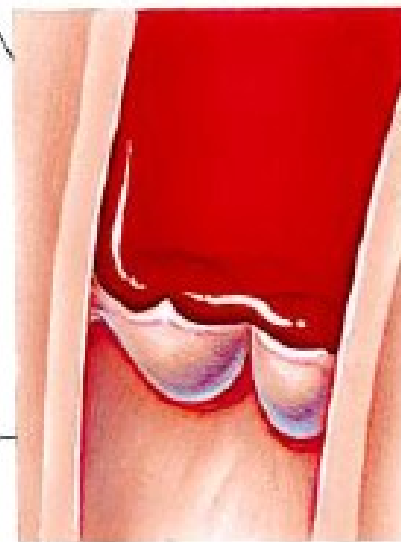
As ventricles contract and intraventricular pressure rises, blood is pushed up against semilunar valves, forcing them open



Semilunar valve open



As ventricles relax and intraventricular pressure falls, blood flows back from arteries, filling the cusps of semilunar valves and forcing them to close



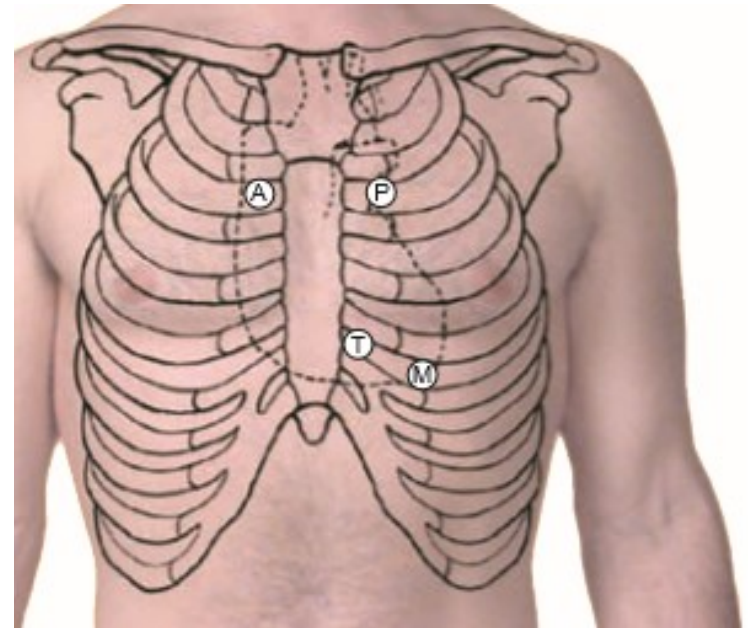
Semilunar valve closed

Heart Sounds

- The **closing** of the **heart valves** causes vibrations in the adjacent blood and heart walls that account for the familiar “**lub-dup**” sounds of the heartbeat
- The “**lub**” is produced by the **closing** of the **AV valves** at the **start of ventricular systole**
- The “**dup**” is produced by the **closing** of the **semilunar valves** at **the end of ventricular systole**
- Sounds are heard away from the valves in the direction of the blood flow

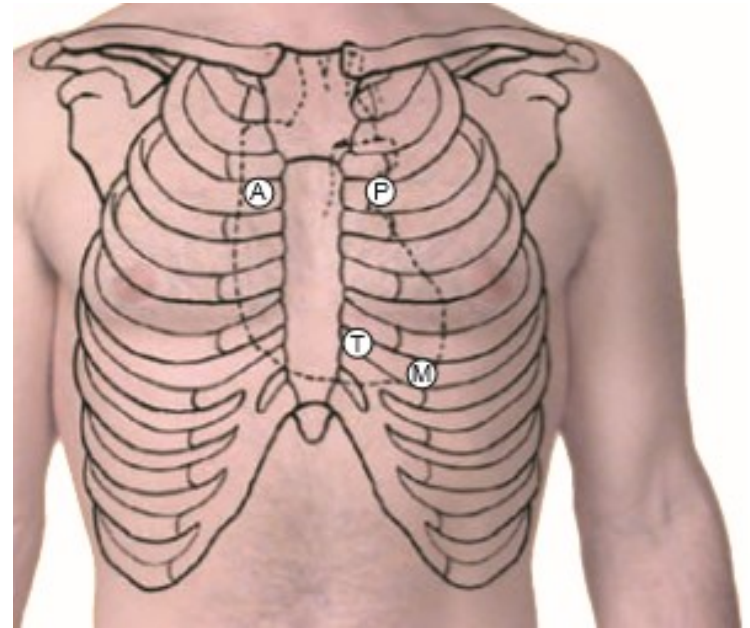
Auscultator areas

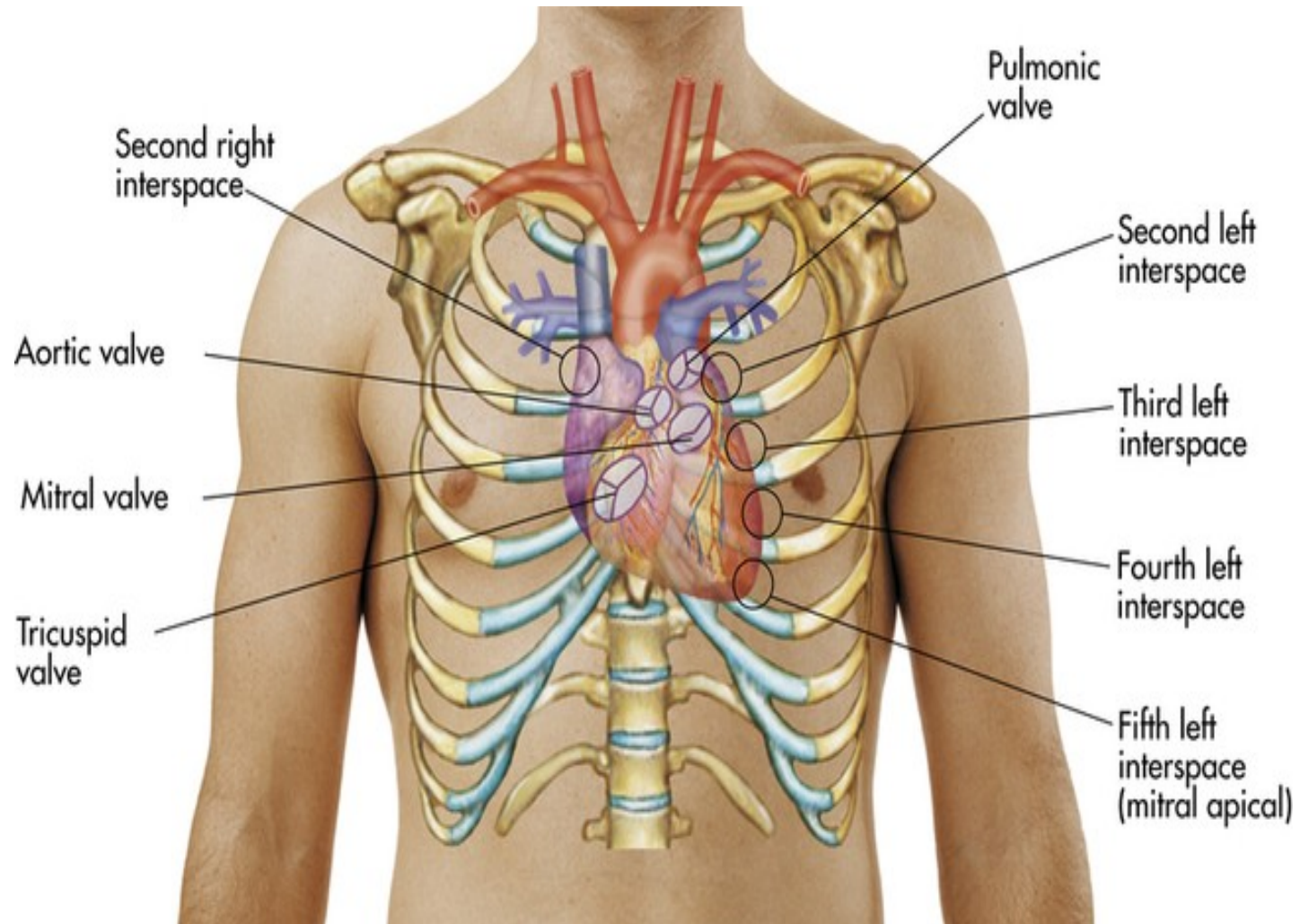
- The **aortic valve** sound is heard over the medial end of the **right 2nd intercostal space**.
- The **pulmonary valve** sound is heard over the medial end of the **left 2nd intercostal space**.



Auscultator areas

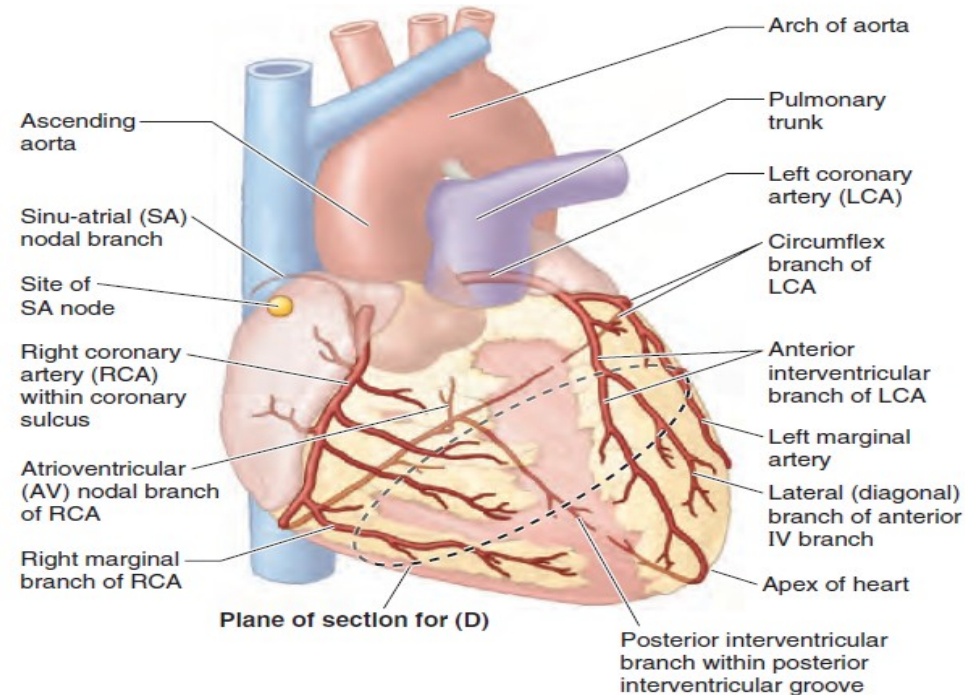
- The sound of the tricuspid valve is heard just to the **Left** margin of the lower part of the sternum at the **4th or 5th intercostal space**.
- The mitral valve is heard over the apex of the heart at the **left 5th intercostal space**





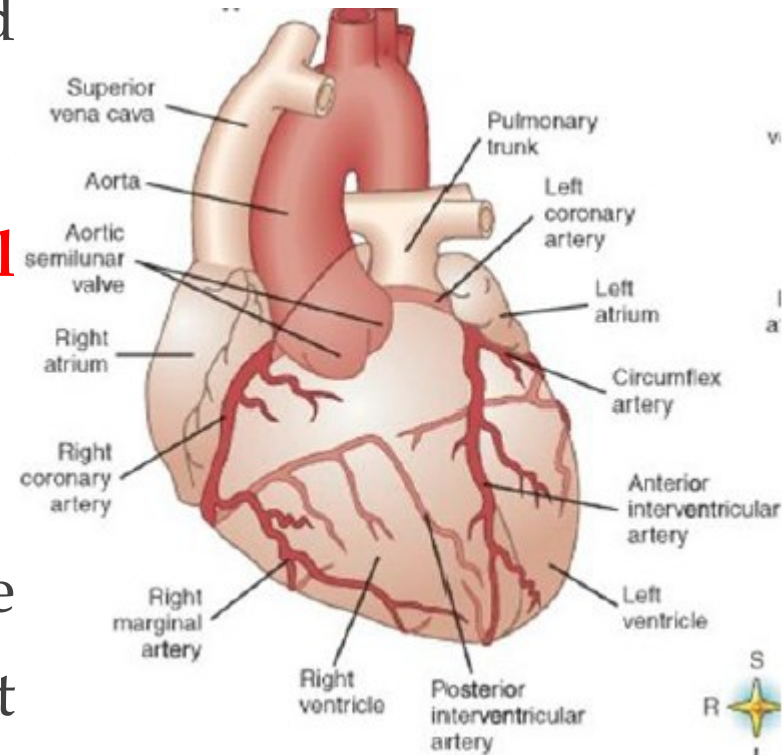
Arterial Supply of Heart

- The **heart** is supplied by two major **coronary arteries**:
 - Right coronary artery &
 - Left coronary artery
- Both arise from Rt and Lt **aortic sinuses**



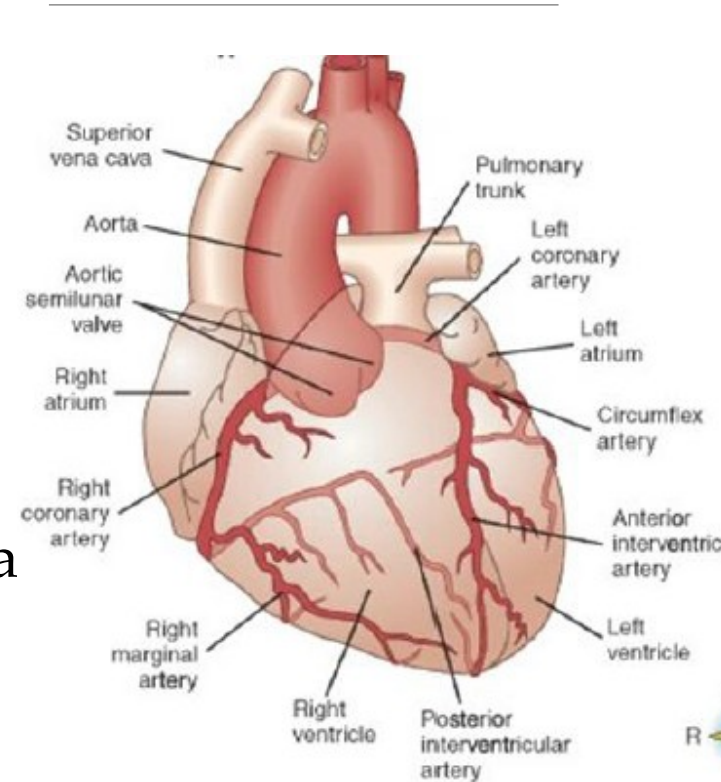
Right coronary artery

- Arise from **right aortic sinus** and runs in coronary groove
- Near its origin gives off **SA nodal** branch
 - supply SA node
- Then courses to the right side of the heart in coronary groove where it gives off the **right marginal branch**
 - supplies the lateral part of the right side of the heart



Right coronary artery

- It turns to the left and continues in **posterior part of coronary groove**
- At crux of heart (the junction of the interatrial and interventricular septa) RCA gives rise to, **AV nodal branch**
 - which supplies the *AV node*
- Then gives off **posterior interventricular branch** that runs in posterior interventricular groove to the apex of the heart
 - Supplies **area of both ventricles** and sends interventricular septal branches



Right coronary artery supplies

The whole of right atrium

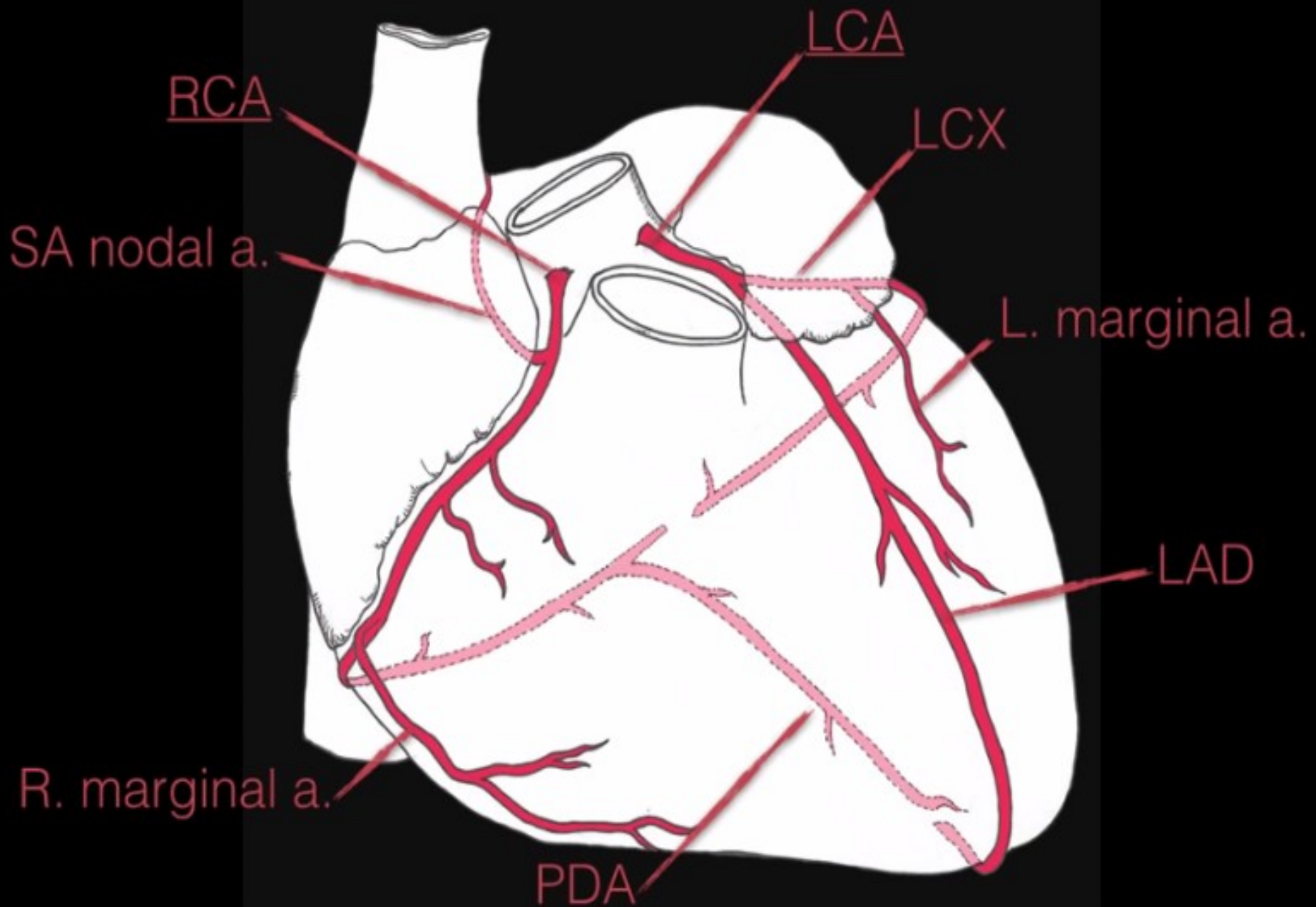
Most of the right ventricle

Part of left ventricle (diaphragmatic surface)

Part of interventricular septum (posterior third)

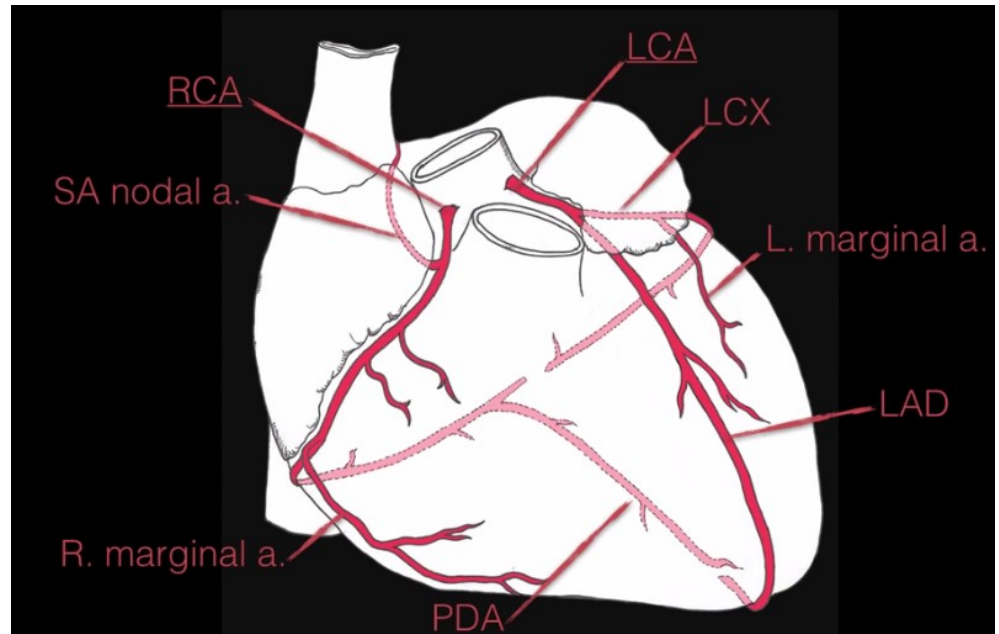
SA node (in 60% of people)

AV node (in 80% of people)



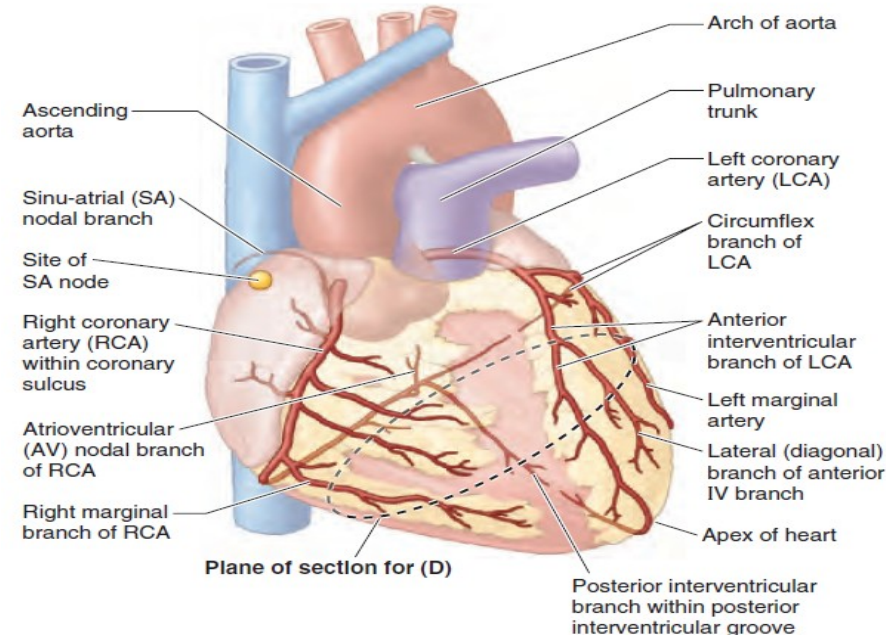
The left coronary artery

- Arise from **left aortic sinus** and passes between left auricle and pulmonary trunk in coronary groove
- At left end of coronary groove divides into:
 - **Anterior interventricular branch and**
 - **Circumflex branch**



The left coronary artery...

- **AIV branch** passes along the interventricular groove to the apex
- It turns around and anastomose with posterior interventricular branch of the right
- **Supplies both ventricles and interventricular septum**
- It also gives lateral branch
- **The circumflex branch** follows coronary groove to posterior surface
 - It gives **left marginal artery** which supplies **left ventricle**
 - In 40% of people SA nodal branch arises from it



The left coronary artery supplies

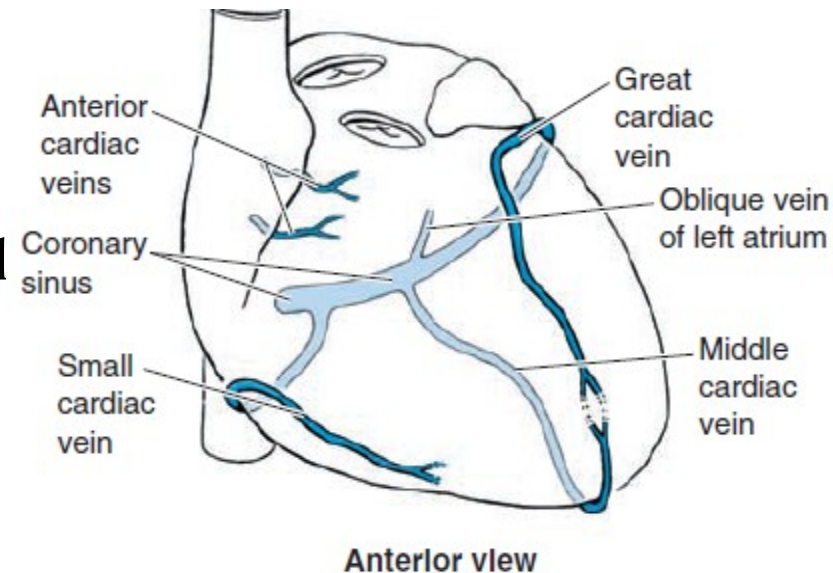
- The left atrium
- Most of the left ventricle
- Part of right ventricle
- Most of interventricular septum (anterior two third)
- SA node (in 40% of people)

Cardiac dominance

- Dominance of coronary arterial system defined by which artery give rise to the PDA
- Variation of coronary dominance includes
 - Right dominance and
 - Left dominance
 - Co-dominance (CD) based on the vascular supply of the posterior interventricular septum (IVS)
- If PDA arises from
 - RCA - Right cardiac dominance (67%)
 - LCA - left cardiac dominance (15%)
 - Both RCA and LCA - codominant in 18%

Venous Drainage of the Heart

- About 2/3rds of the venous drainage of the heart is by veins which accompany the arteries
- And drain into the **coronary sinus** in the coronary sulcus, located posteriorly



i) Great cardiac vein

- ✓ is the main tributary of the coronary sinus
- ✓ drains the areas of the heart supplied by the LCA.
- ✓ accompanies **anterior interventricular artery (AIVA) (LDA)**

Venous Drainage of the Heart

ii) Middle cardiac vein

- ✓ accompanies PIVA

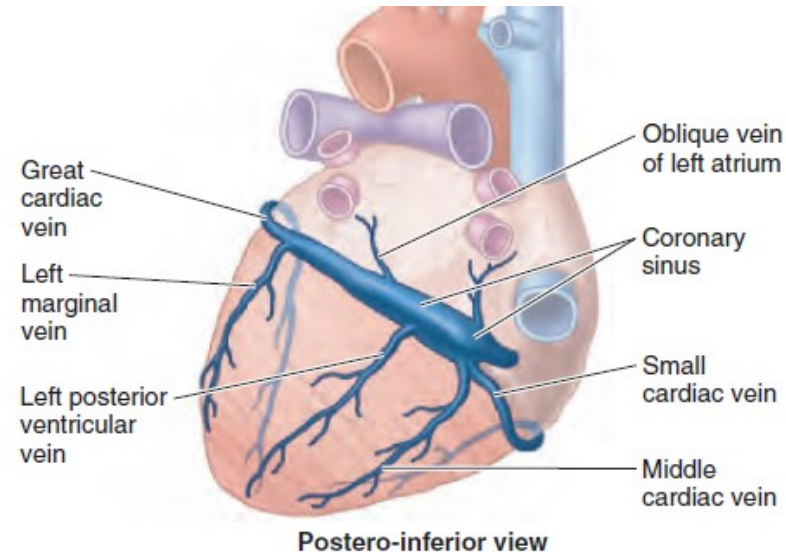
iii) Small cardiac vein

- ✓ accompanies right marginal artery

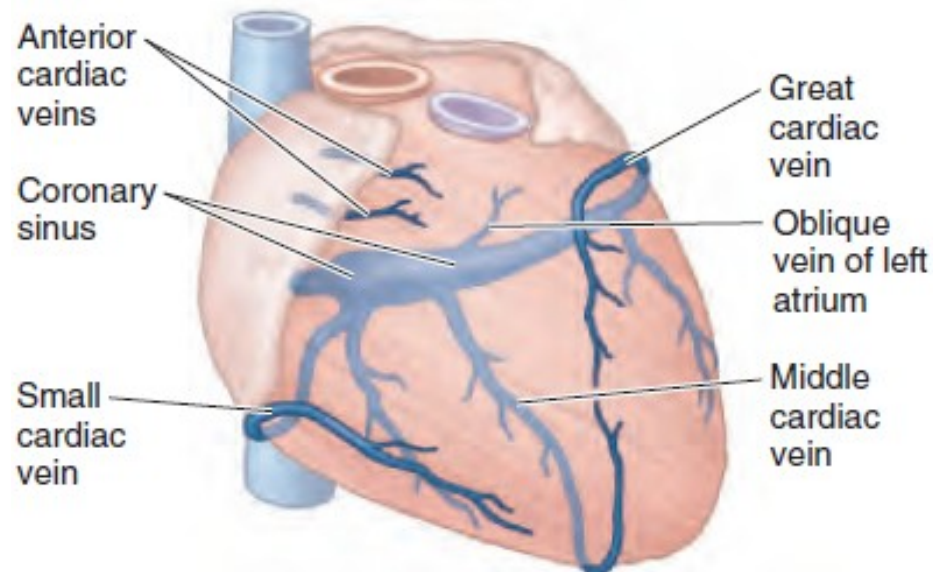
➤ Thus these two veins drain most of the areas commonly supplied by the RCA

iv) Oblique vein of left atrium:

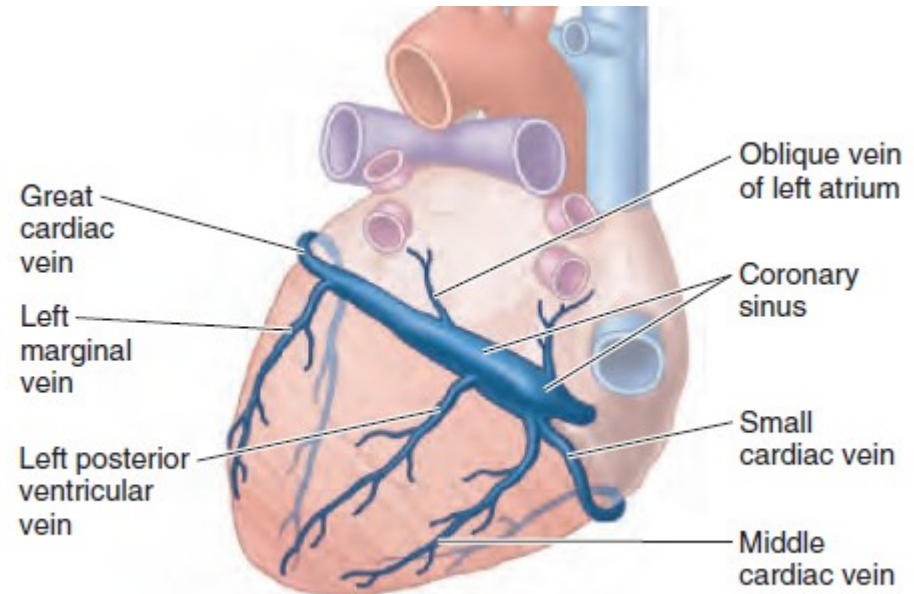
- ✓ descends over the posterior wall of the **left atrium** and merges with the great cardiac vein to form the **coronary sinus**



Venous Drainage of the Heart...



Anterior view



Postero-inferior view

Innervation of the heart

- The heart is supplied by autonomic nerve fibers from the **cardiac plexus**
- Which is often divided into **superficial** and **deep portions**.
- This nerve network is most commonly described as lying on the anterior surface of the **bifurcation of the trachea**
- The **cardiac plexus** is formed of both **sympathetic** and **parasympathetic fibers**, as well as **visceral afferent fibers**

Innervation of the heart

Parasympathetic nerve supply : arise as branches of the **Vagus nerve**

- Parasympathetic stimulation **slows the heart rate**, reduces force of contraction and **constricts coronary arteries**

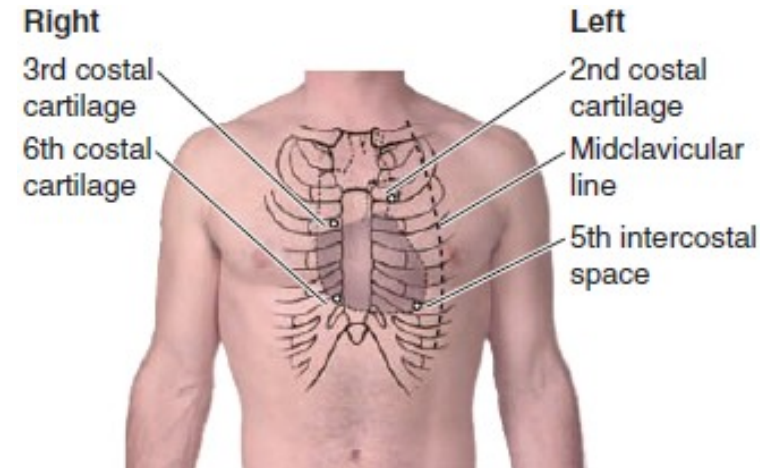
Sympathetic nerves : **cervical** and **upper thoracic chain ganglia**

- Sympathetic stimulation **increases the heart rate** and **force of heart contractions** and produce **dilation of coronary arteries**

Surface Anatomy of the Heart

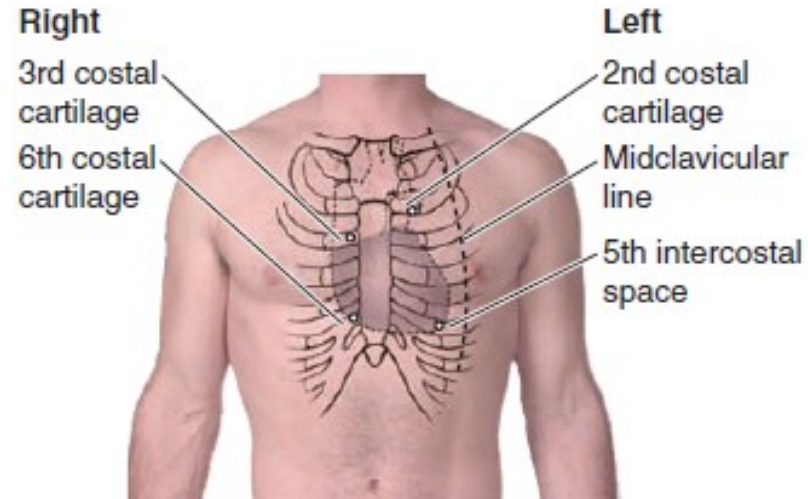
➤ The outline of the heart can be traced on the anterior surface of the thorax by using the following guidelines

- ✓ The **superior border is** formed by root of great vessels. corresponds to a line connecting the inferior border of the **2nd left costal cartilage** to the superior border of the 3rd right costal cartilage.
- ✓ The **right border** corresponds to a line drawn from the 3rd right costal cartilage to the 6th right costal cartilage; this border is slightly convex to the right.



Surface Anatomy of the Heart...

- ✓ The *inferior border* corresponds to a line drawn from the inferior end of the right border to a point in the 5th intercostal space close to the left MCL; the left end of this line corresponds to the location of the **apex** of the heart.
- ✓ The *left border* corresponds to a line connecting the left ends of the lines representing the superior and inferior borders.



Surface Anatomy of the Heart...

- ✓ The **valves** are located posterior to the sternum
- ✓ however, the sounds produced by them are projected to the areas shown in pulmonary (*P*), aortic (*A*), mitral (*M*), and tricuspid (*T*) where the stethoscope may be placed to avoid intervening bone.

